Diffusion models

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Цель

- Познакомится с концептом диффузионных моделей, рассмотреть различные практические применения и расширения диффузионных моделей

План

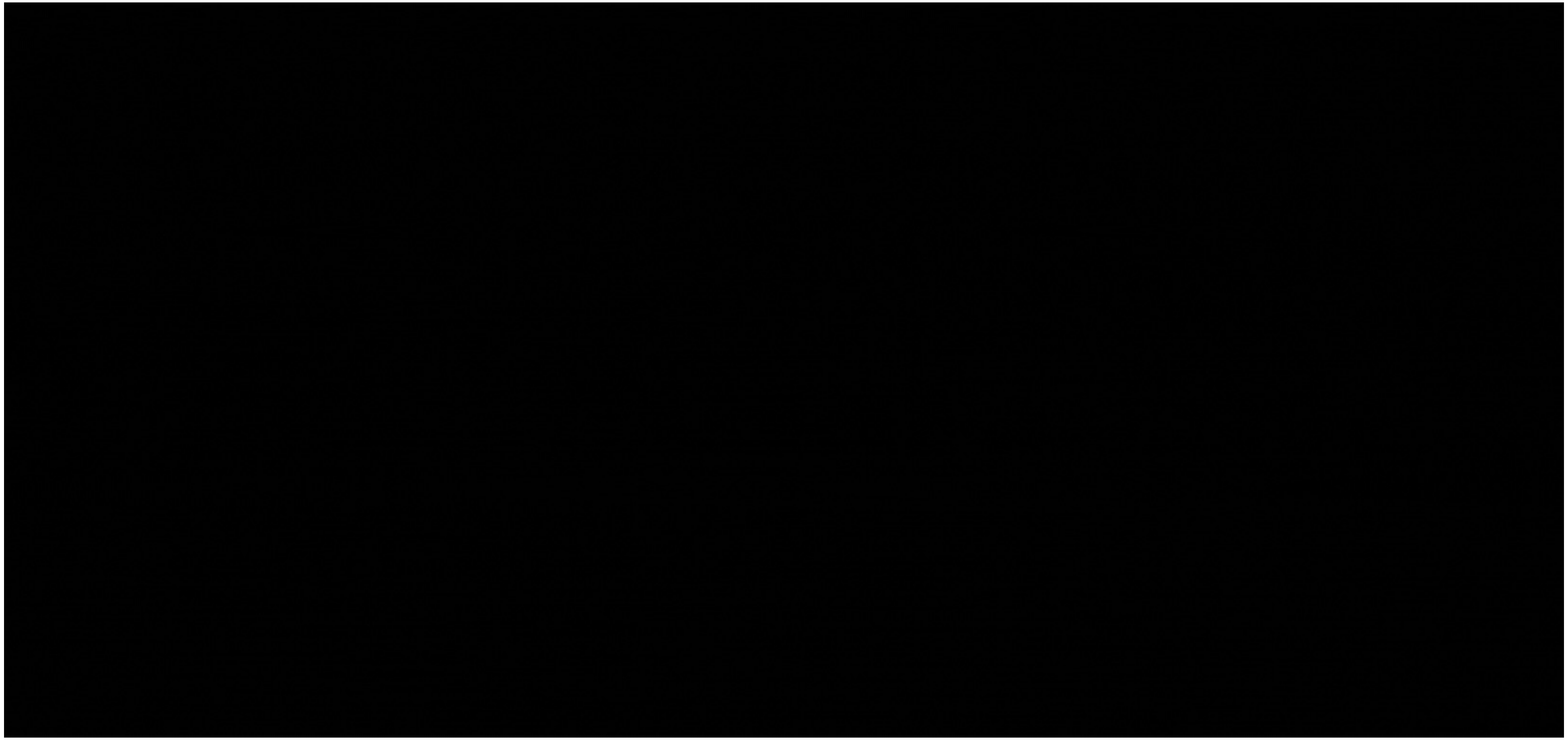
- Прямой и обратный диффузионные процессы (DDPM)
- Условная генерация с помощью диффузионных моделей
- Latent Diffusion Models и Stable Diffusion
- Применения диффузионных моделей
- Методы дообучения диффузионных моделей
- Методы добавления новых условий в обученные модели
- Инструменты для работы с диффузионными моделями

Прямой и обратный диффузионные процессы (DDPM)

Denoising Diffusion Probabilistic Models

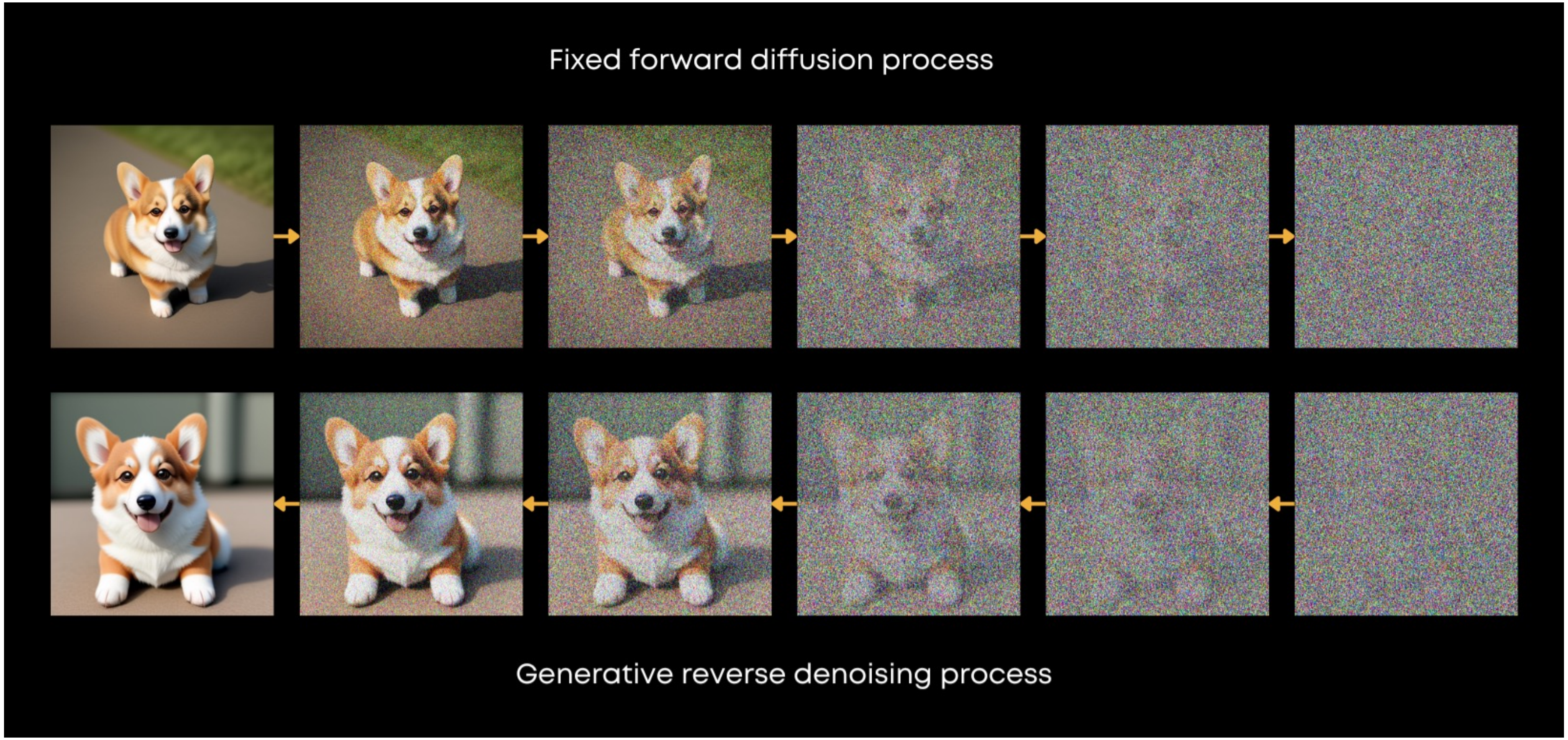
[Denoising Diffusion Probabilistic Models, Arxiv](#)

Прямой и обратный диффузионные процессы



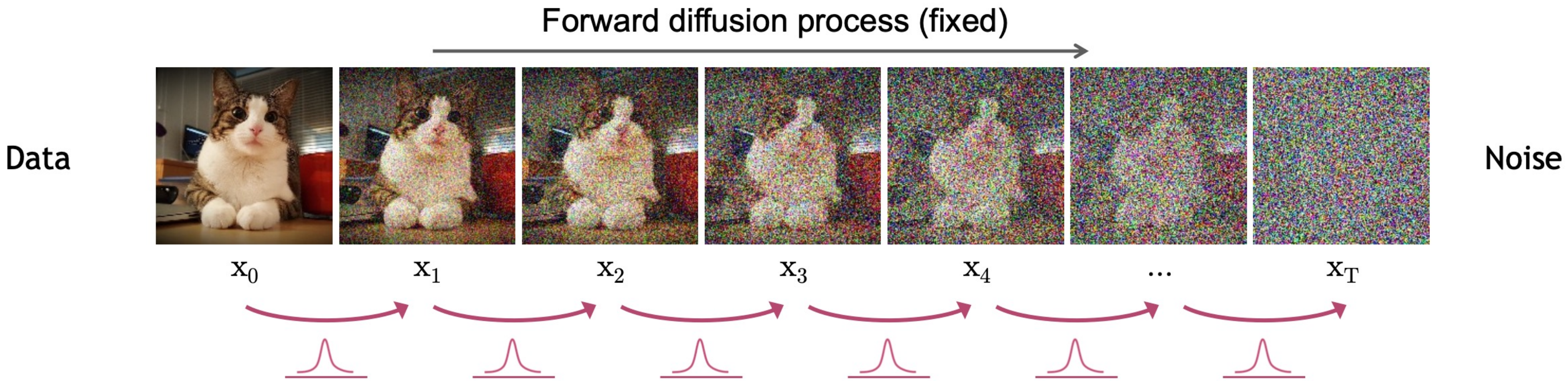
Denoising Diffusion Probabilistic Models, Arxiv

Прямой и обратный диффузионные процессы



[Denoising Diffusion Probabilistic Models, Arxiv]

Прямой диффузионный процесс



- В теории:

$$x_0 \Rightarrow x_1 \Rightarrow \cdots \Rightarrow x_T$$

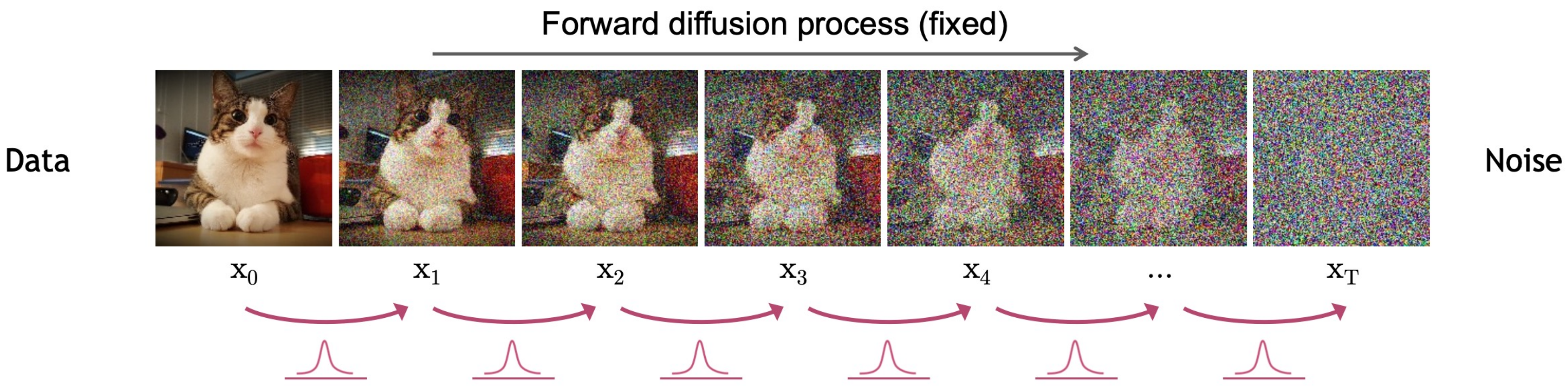
$$q(x_t|x_{t-1}) = \mathbb{N}\left(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t \mathbb{I}\right); q(x_T|x_0) = \mathbb{N}(x_t; 0{,}1)$$

$$\beta_0 = 0, \beta_T = 1, \beta_t \underset{T}{>} \beta_{t-1}, T = 1000$$

$$q(x_{1:T}|x_0) = \prod_{t=1}^{T} q(x_t|x_{t-1})$$

CVPR 2022 Tutorial on Diffusion Models

Прямой диффузионный процесс



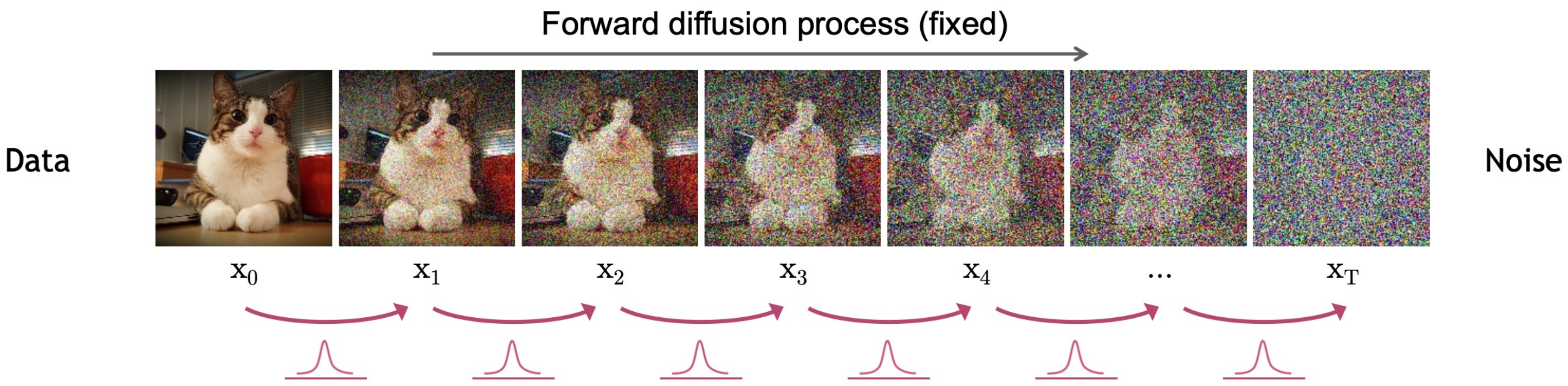
- На практике:

$$\epsilon \sim \mathbb{N}(0,1)$$

$$x_t = \sqrt{1-\beta_t} * x_{t-1} + \sqrt{\beta_t} * \epsilon$$

[CVPR 2022 Tutorial on Diffusion Models]

Прямой диффузионный процесс



- На практике:

$$\epsilon \sim \mathbb{N}(0,1)$$

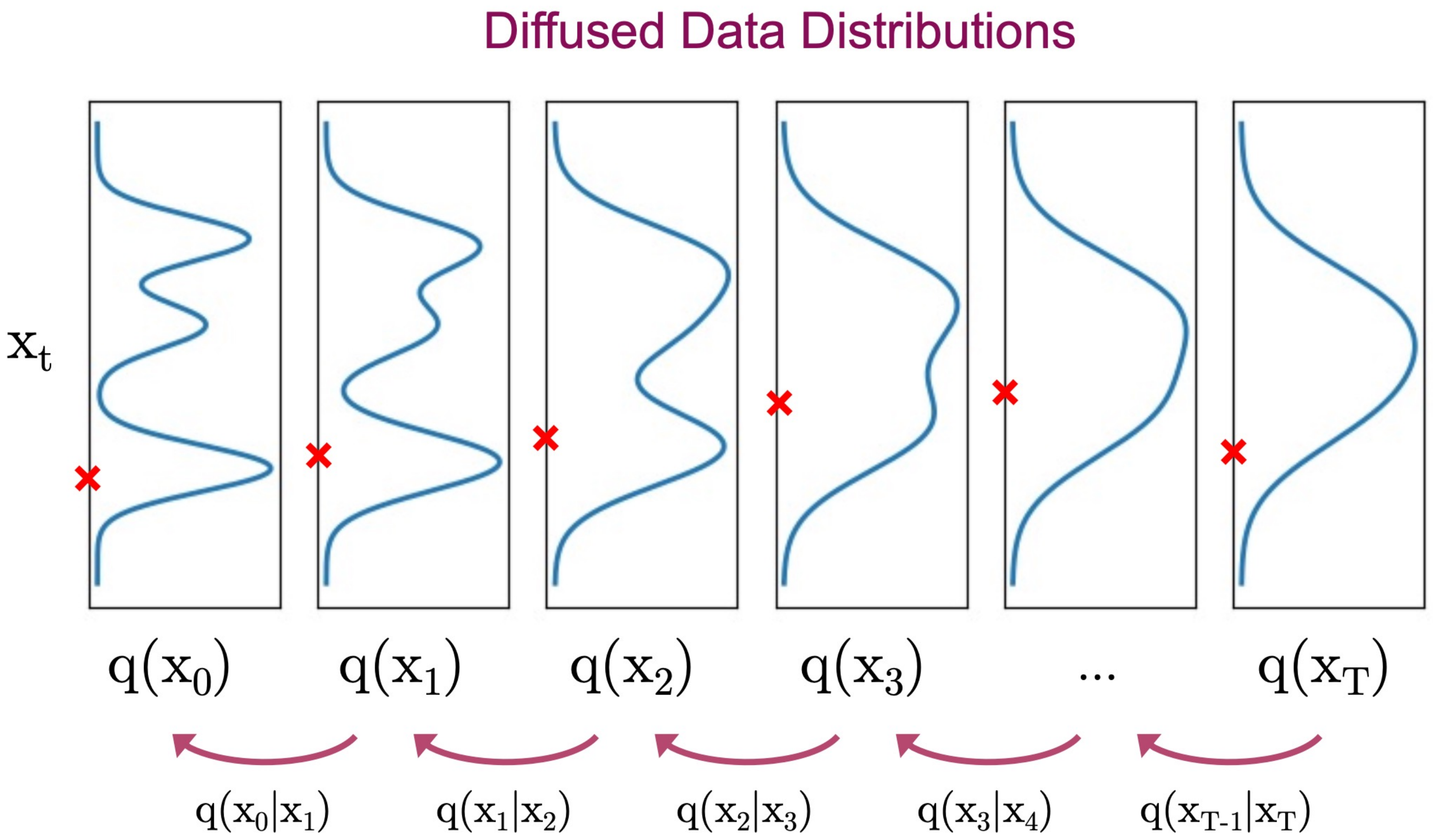
$$\text{введя } \alpha_t = 1 - \beta_t, \bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$$

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} * \epsilon$$

$$\beta_t \text{ выбираются так, чтобы } \bar{\alpha}_t \to 0 \text{ и } q(x_T | x_0) \approx N(x_T; 0,1)$$

[CVPR 2022 Tutorial on Diffusion Models]

Обратный диффузионный процесс



Можем ли мы апроксимировать $q(x_{t-1}|x_t)$?

Да, мы можем использовать **нормальное распределение**, при условии, что изменение β_t малое на каждом прямом диффузионном шаге.

На самом деле описать $q(x_{t-1}|x_t, x_0)$ достаточно легко, об этом далее.

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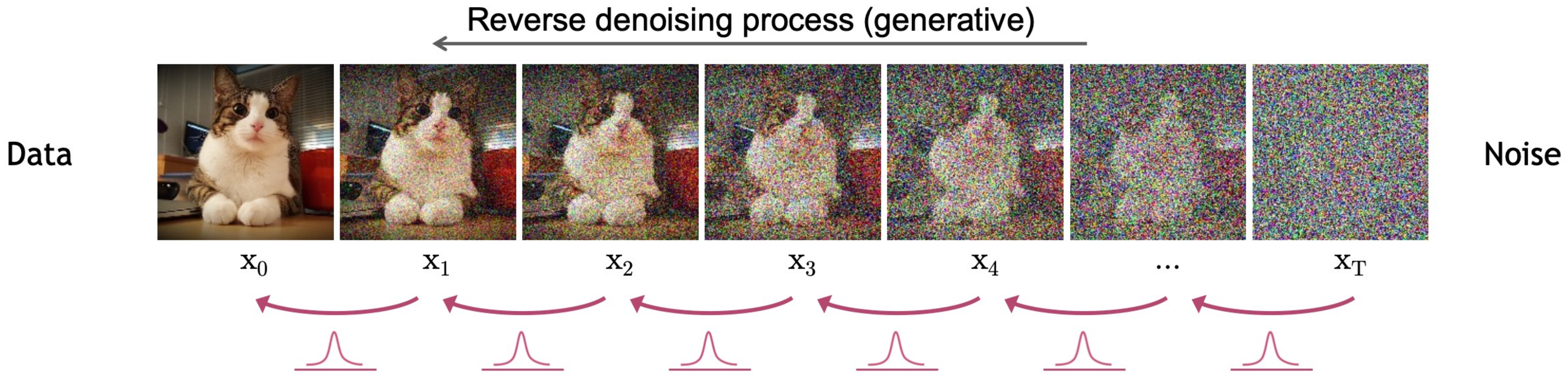
Обратный диффузионный процесс



$$p(\mathbf{x}_T) = \mathcal{N}(\mathbf{x}_T; \mathbf{0}, \mathbf{I})$$

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \underbrace{\mu_\theta(\mathbf{x}_t, t)}, \sigma_t^2\mathbf{I})$$

$$\Rightarrow \quad p_\theta(\mathbf{x}_{0:T}) = p(\mathbf{x}_T)\prod_{t=1}^{T} p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$$

Trainable network
(U-net, Denoising Autoencoder)

[CVPR 2022 Tutorial on Diffusion Models]

Обратный диффузионный процесс
Вариационная нижняя оценка

$$\mathbb{E}_{q(\mathbf{x}_0)}\left[-\log p_\theta(\mathbf{x}_0)\right] \leq \mathbb{E}_{q(\mathbf{x}_0)q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\right] =: L$$

$$L = \mathbb{E}_q\left[\underbrace{D_{\mathrm{KL}}(q(\mathbf{x}_T|\mathbf{x}_0)||p(\mathbf{x}_T))}_{L_T} + \sum_{t>1}\underbrace{D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)||p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{L_{t-1}} \underbrace{- \log p_\theta(\mathbf{x}_0|\mathbf{x}_1)}_{L_0}\right]$$

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1};\tilde{\mu}_t(\mathbf{x}_t,\mathbf{x}_0),\tilde{\beta}_t\mathbf{I}),$$

$$\text{where}\quad \tilde{\mu}_t(\mathbf{x}_t,\mathbf{x}_0) := \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}\mathbf{x}_0 + \frac{\sqrt{1-\beta_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\mathbf{x}_t \quad\text{and}\quad \tilde{\beta}_t := \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t$$

[Denoising Diffusion Probabilistic Models, Arxiv]

Обратный диффузионный процесс
Вариационная нижняя оценка

- Поскольку мы предполагаем, что $q(x_{t-1}|x_t, x_0)$ и $p_\theta(x_{t-1}|x_t)$ - нормальные распределения:

$$L_{t-1} = D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)||p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)) = \mathbb{E}_q\left[\frac{1}{2\sigma_t^2}||\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) - \mu_\theta(\mathbf{x}_t, t)||^2\right] + C$$

$$\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) = \frac{1}{\sqrt{1-\beta_t}}\left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon\right) \quad \mu_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{1-\beta_t}}\left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(\mathbf{x}_t, t)\right)$$

$$L_{t-1} = \mathbb{E}_{\mathbf{x}_0 \sim q(\mathbf{x}_0), \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})}\left[\frac{\beta_t^2}{2\sigma_t^2(1-\beta_t)(1-\bar{\alpha}_t)}||\epsilon - \epsilon_\theta(\underbrace{\sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\epsilon}_{\mathbf{x}_t}, t)||^2\right] + C$$

[Denoising Diffusion Probabilistic Models, Arxiv]()

Обратный диффузионный процесс
Вариационная нижняя оценка

$$L_{t-1} = \mathbb{E}_{\mathbf{x}_0 \sim q(\mathbf{x}_0), \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} \left[\underbrace{\frac{\beta_t^2}{2\sigma_t^2 (1-\beta_t)(1-\bar{\alpha}_t)}}_{\lambda_t} ||\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t}\, \mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon, t)||^2 \right]$$

$$L_{\text{simple}} = \mathbb{E}_{\mathbf{x}_0 \sim q(\mathbf{x}_0), \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), t \sim \mathcal{U}(1,T)} \left[||\epsilon - \epsilon_\theta(\underbrace{\sqrt{\bar{\alpha}_t}\, \mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon}_{\mathbf{x}_t}, t)||^2 \right]$$

[Denoising Diffusion Probabilistic Models, Arxiv]()

Расписание шума
Косинусное и линейное расписания

$$\bar{\alpha}_t = \frac{f(t)}{f(0)}, f(t) = \cos\left(\frac{\frac{t}{T} + s}{1 + s} * \frac{\pi}{2}\right)^2$$

$$\beta_t = 1 - \frac{\bar{\alpha}_t}{\bar{\alpha}_{t-1}}$$

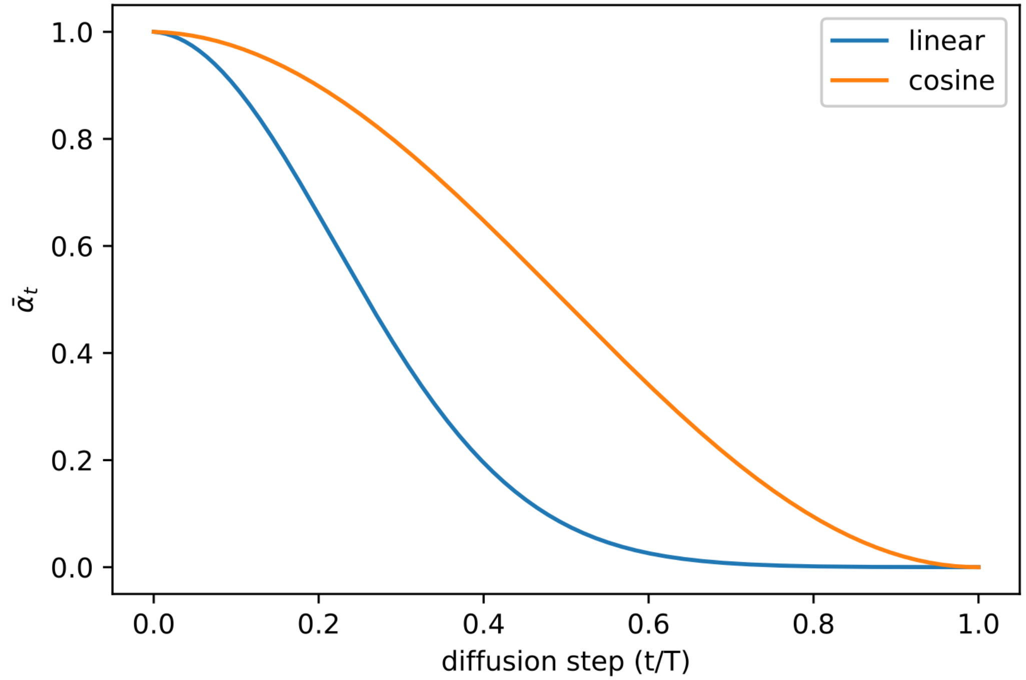


Figure 5. $\bar{\alpha}_t$ throughout diffusion in the linear schedule and our proposed cosine schedule.

[Improved Denoising Diffusion Probabilistic Models]()

Расписание шума
Косинусное и линейное расписания

Figure 3. Latent samples from linear (top) and cosine (bottom) schedules respectively at linearly spaced values of t from 0 to T. The latents in the last quarter of the linear schedule are almost purely noise, whereas the cosine schedule adds noise more slowly

Improved Denoising Diffusion Probabilistic Models

Процесс обучения и применения диффузионной модели

Algorithm 1 Training

1: **repeat**
2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
3: $t \sim \mathrm{Uniform}(\{1, \ldots, T\})$
4: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
5: Take gradient descent step on
$$\nabla_\theta \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t) \right\|^2$$
6: **until** converged

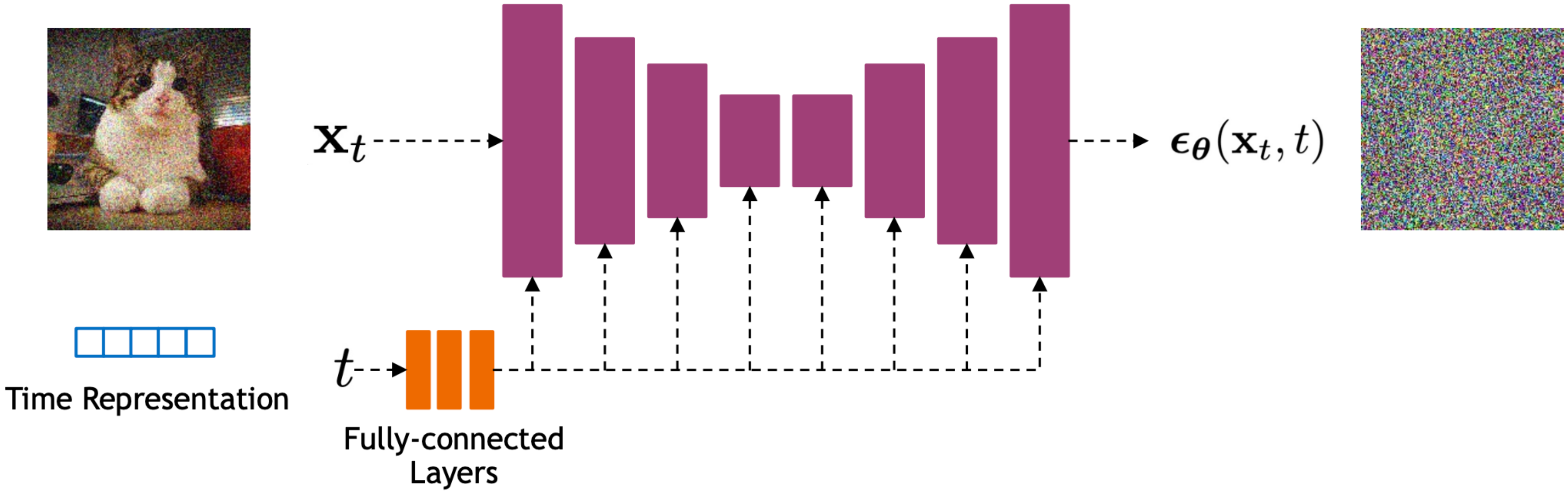
Algorithm 2 Sampling

1: $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
2: **for** $t = T, \ldots, 1$ **do**
3: $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
4: $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$
5: **end for**
6: **return** $\mathbf{x}_0$

[Denoising Diffusion Probabilistic Models, Arxiv]

Условная генерация с помощью диффузионных моделей

Генерация без условия



[CVPR 2022 Tutorial on Diffusion Models]

Условная генерация

- Очевидно, что мы можем переучить модель с нуля, добавив в архитектуру механизмы, которые смогут получать и обрабатывать дополнительный сигнал, например эмбеддинг текста или картинки. Удобнее всего использовать cross-attention и блоки модели-трансформера

- Обучение новой модели – долгий и дорогостоящий процесс, можно ли обойтись без переобучения?

Classifier-guidance

- Пусть $p_\phi(y|x_t)$ - классификатор, обученный на зашумленных объектах, можем использовать его градиенты в качестве поправки к оценке предсказанного нашей моделью шума

Algorithm 2 Classifier guided DDIM sampling, given a diffusion model $\epsilon_\theta(x_t)$, classifier $p_\phi(y|x_t)$, and gradient scale s.

Input: class label y, gradient scale s
$x_T \leftarrow$ sample from $\mathcal{N}(0, \mathbf{I})$
for all t from T to 1 **do**
  $\hat{\epsilon} \leftarrow \epsilon_\theta(x_t) - \sqrt{1-\bar{\alpha}_t}\, \nabla_{x_t} \log p_\phi(y|x_t)$
  $x_{t-1} \leftarrow \sqrt{\bar{\alpha}_{t-1}} \left(\frac{x_t - \sqrt{1-\bar{\alpha}_t}\hat{\epsilon}}{\sqrt{\bar{\alpha}_t}} \right) + \sqrt{1-\bar{\alpha}_{t-1}}\hat{\epsilon}$
end for
return x_0

[Diffusion Models Beat GANs on Image Synthesis, Arxiv]()

Classifier-guidance

Figure 3: Samples from an unconditional diffusion model with classifier guidance to condition on the class "Pembroke Welsh corgi". Using classifier scale 1.0 (left; FID: 33.0) does not produce convincing samples in this class, whereas classifier scale 10.0 (right; FID: 12.0) produces much more class-consistent images.

[Diffusion Models Beat GANs on Image Synthesis, Arxiv]

Classifier-free guidance

- Переобучим модель и добавим в качестве дополнительного входа в сеть, предсказывающую шум, метку класса / эмбеддинг текста или картинки $\epsilon = \epsilon(x_t, t, y)$. Также будем с некоторой вероятностью подавать на вход пустое условие, чтобы модель могла генерировать без условия $\epsilon = \epsilon(x_t, t)$

$$\begin{aligned}
\nabla_{\mathbf{x}_t} \log p(y|\mathbf{x}_t) &= \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t|y) - \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t) \\
&= -\frac{1}{\sqrt{1-\bar{\alpha}_t}} \Big(\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, y) - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \Big) \\
\bar{\boldsymbol{\epsilon}}_\theta(\mathbf{x}_t, t, y) &= \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, y) - \sqrt{1-\bar{\alpha}_t}\, w \nabla_{\mathbf{x}_t} \log p(y|\mathbf{x}_t) \\
&= \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, y) + w\big(\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, y) - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\big) \\
&= (w+1)\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, y) - w\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)
\end{aligned}$$

[Classifier-Free Diffusion Guidance, Arxiv]

Classifier-free guidance

Non-guidance

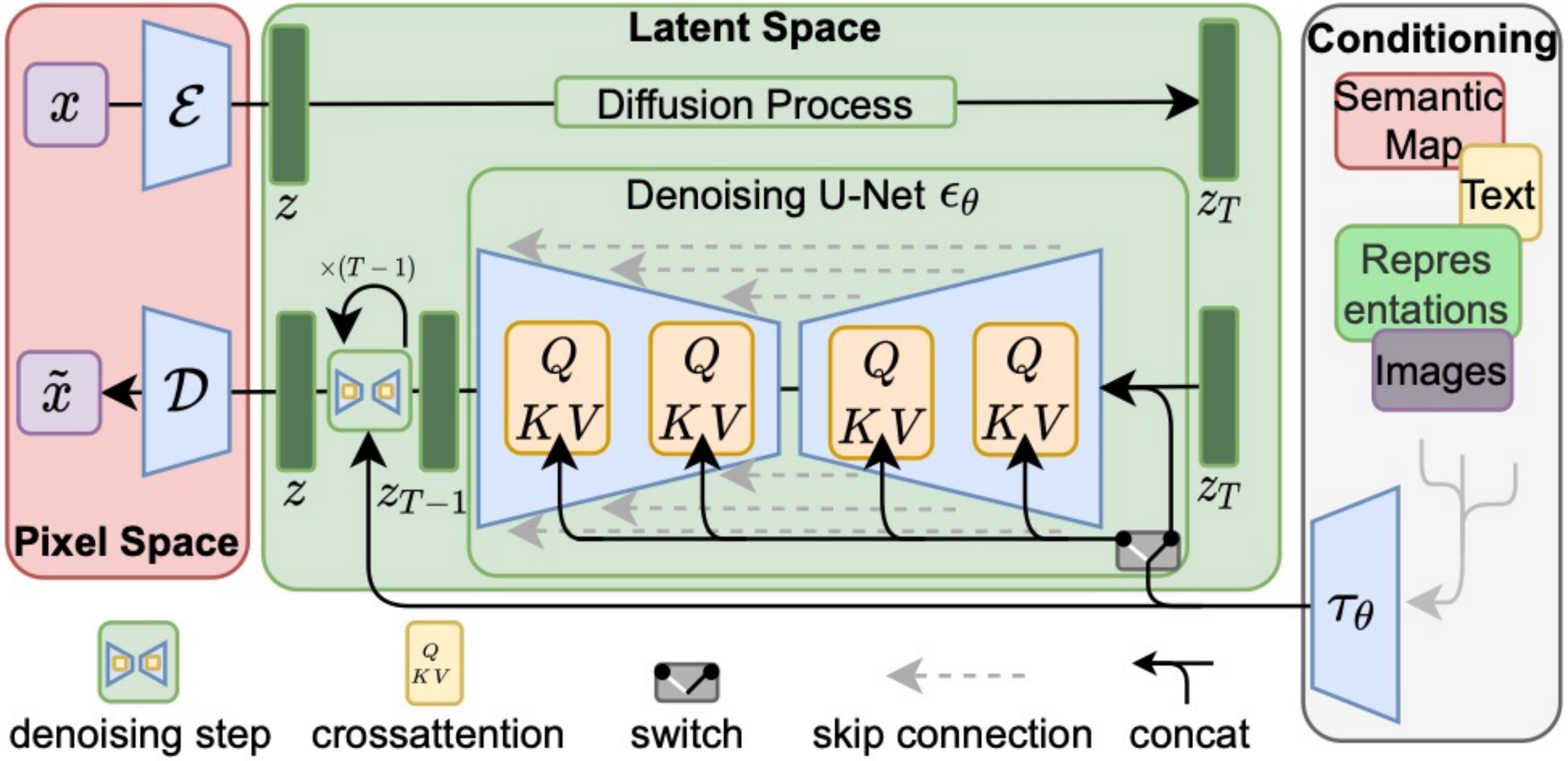
$\omega = 1$

$\omega = 3$

[Classifier-Free Diffusion Guidance, Arxiv]

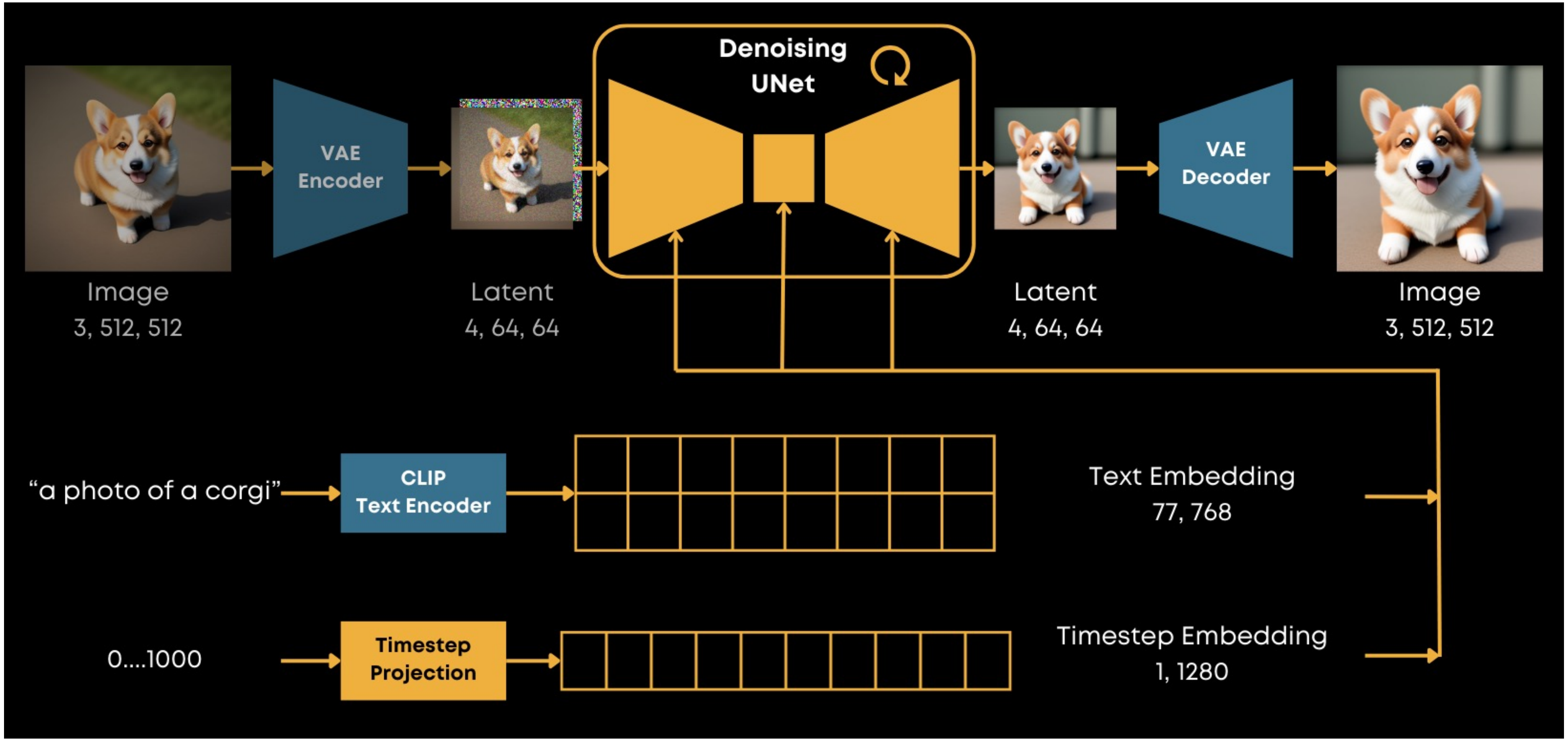
Latent Diffusion Models и Stable Diffusion

Latent Diffusion Models (LDM)



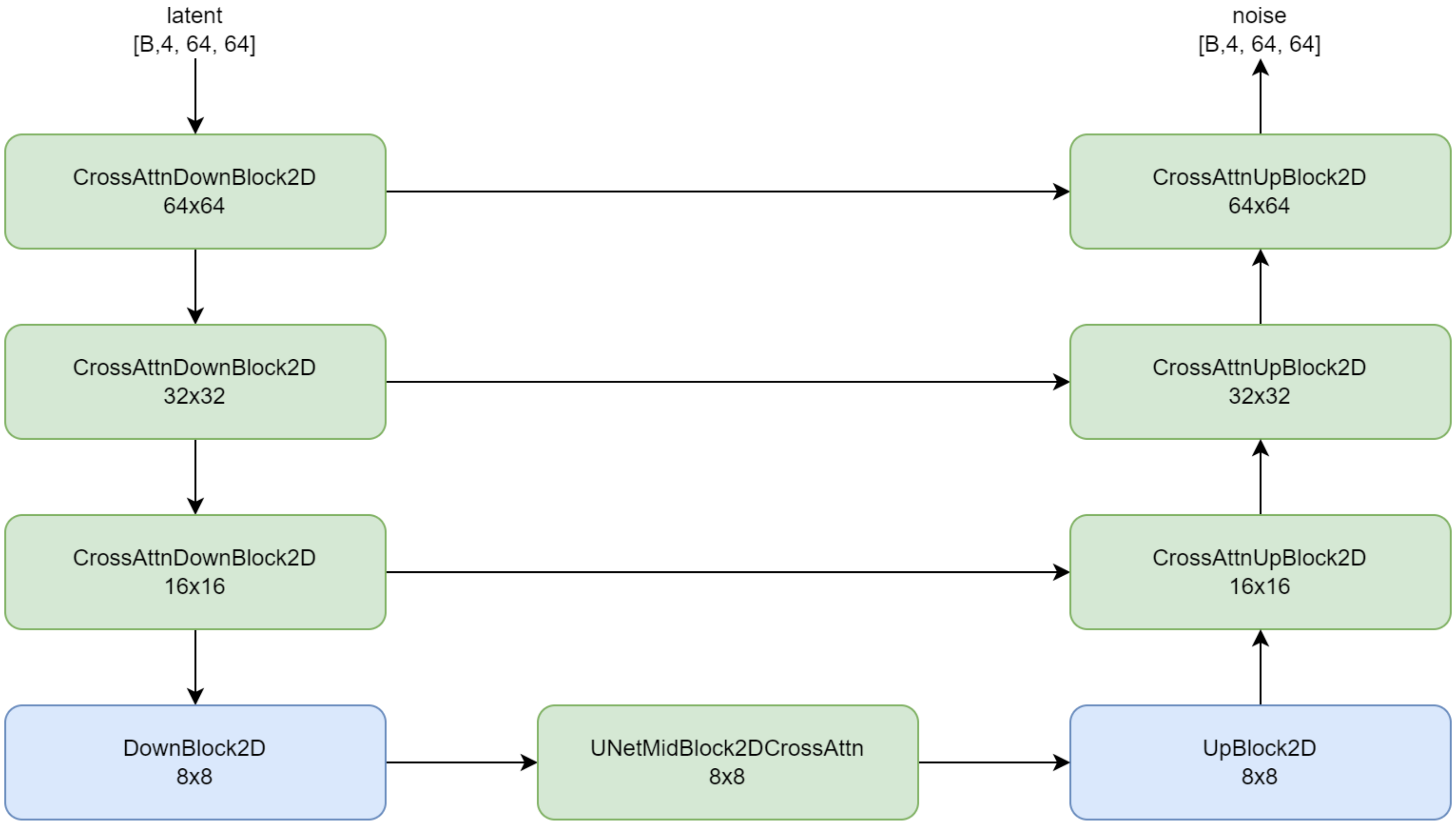
[High-Resolution Image Synthesis with Latent Diffusion Models, Arxiv]

Stable Diffusion (1.5)

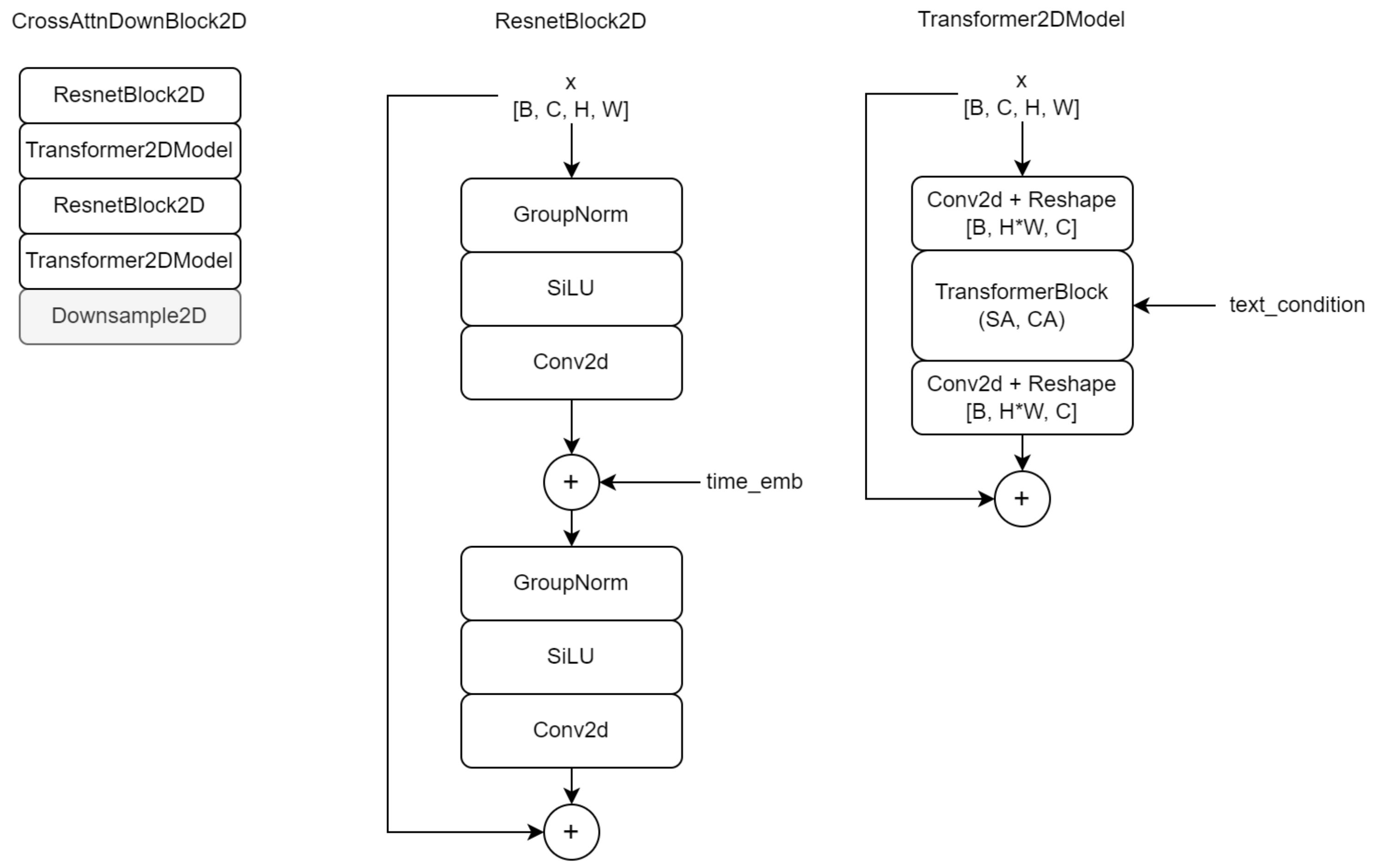


[High-Resolution Image Synthesis with Latent Diffusion Models, Arxiv]

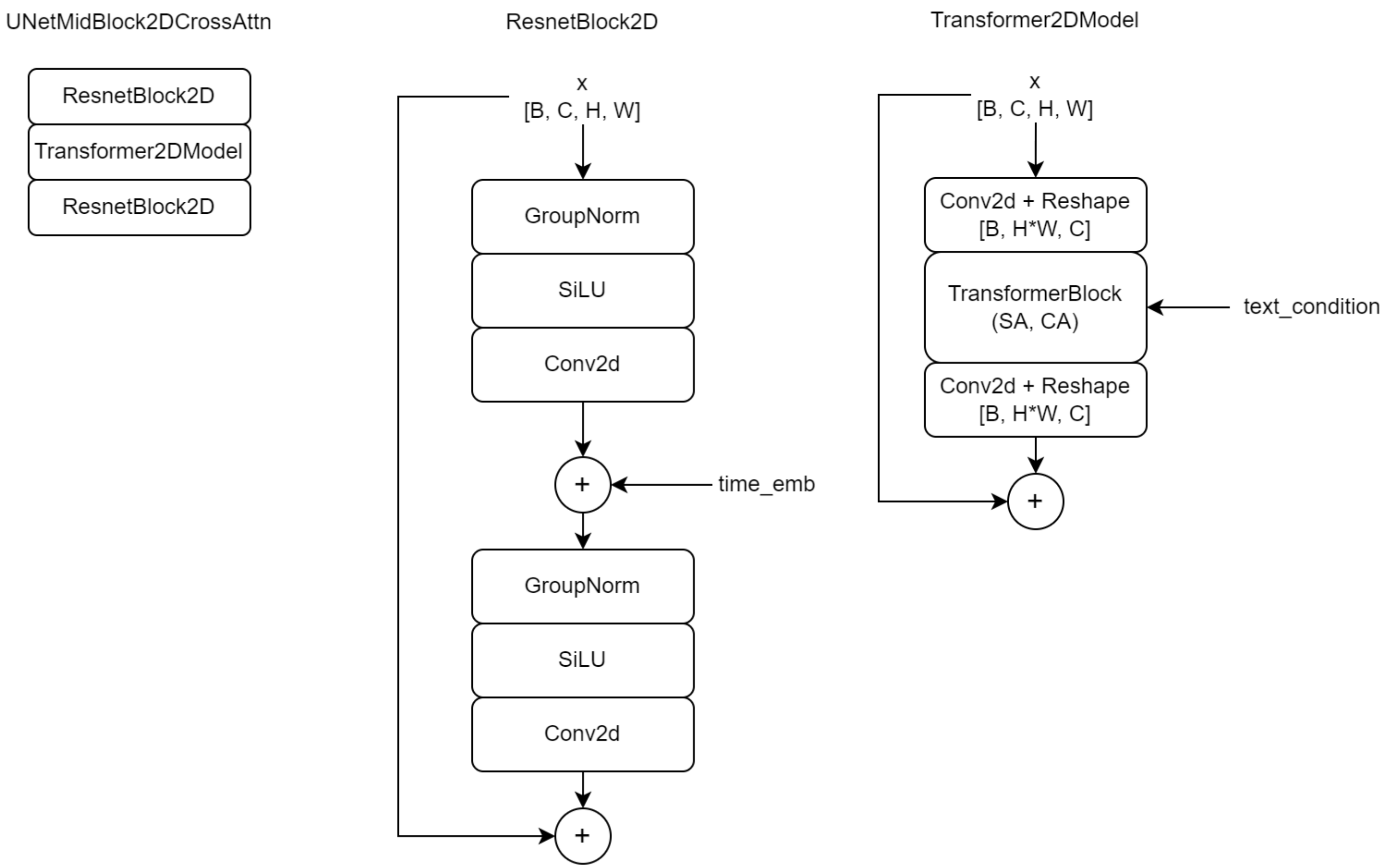
Stable Diffusion (1.5)



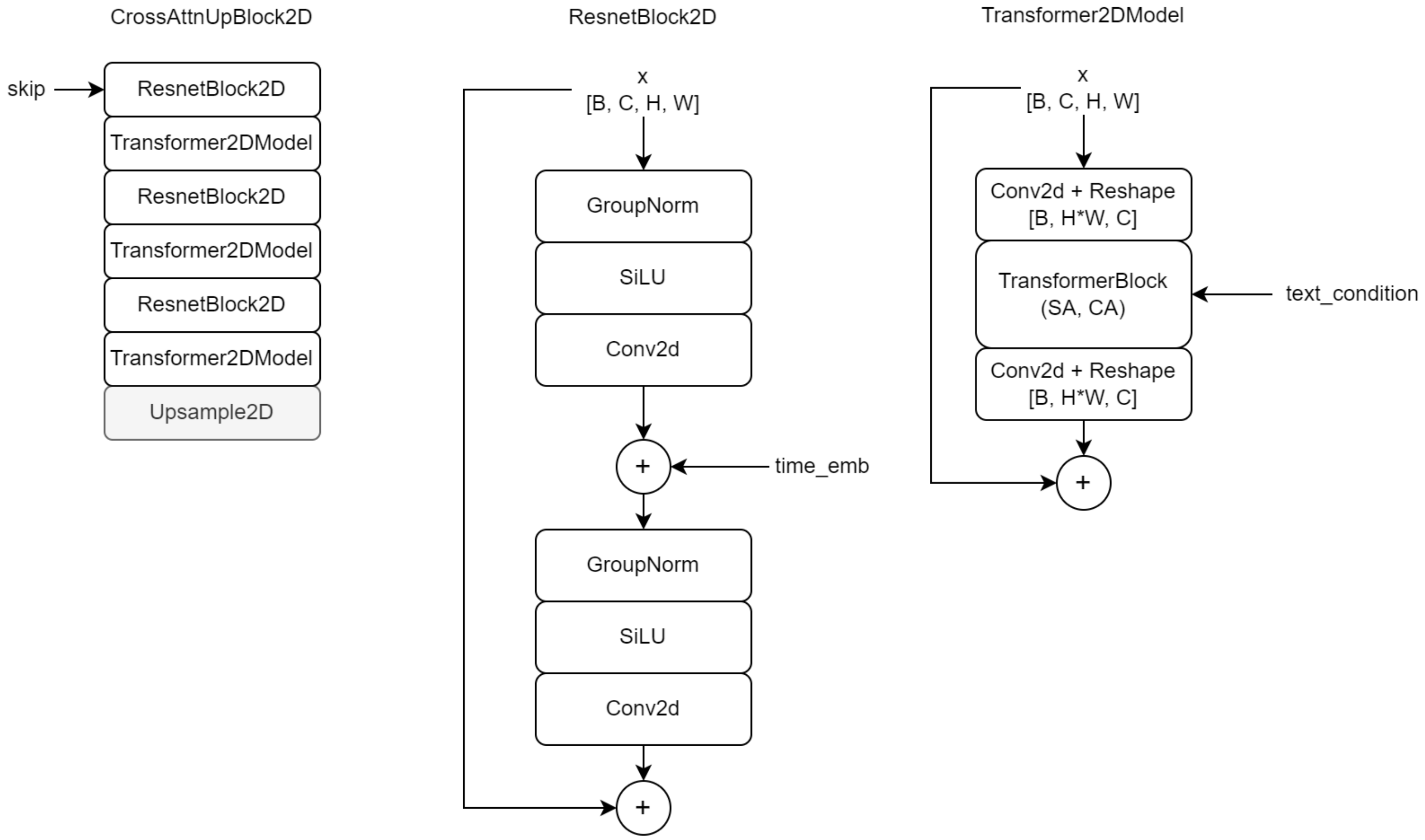
Stable Diffusion (1.5)
CrossAttnDownBlock2D



Stable Diffusion (1.5)
UNetMidBlock2DCrossAttn



Stable Diffusion (1.5)
CrossAttnUpBlock2D



Применения диффузионных моделей

Text-to-image

vibrant portrait painting of Salvador Dalí with a robotic half face

a shiba inu wearing a beret and black turtleneck

a close up of a handpalm with leaves growing from it

an espresso machine that makes coffee from human souls, artstation

panda mad scientist mixing sparkling chemicals, artstation

a corgi's head depicted as an explosion of a nebula

[Hierarchical Text-Conditional Image Generation with CLIP Latents, Arxiv]

Image Inpainting

Image Inpainting

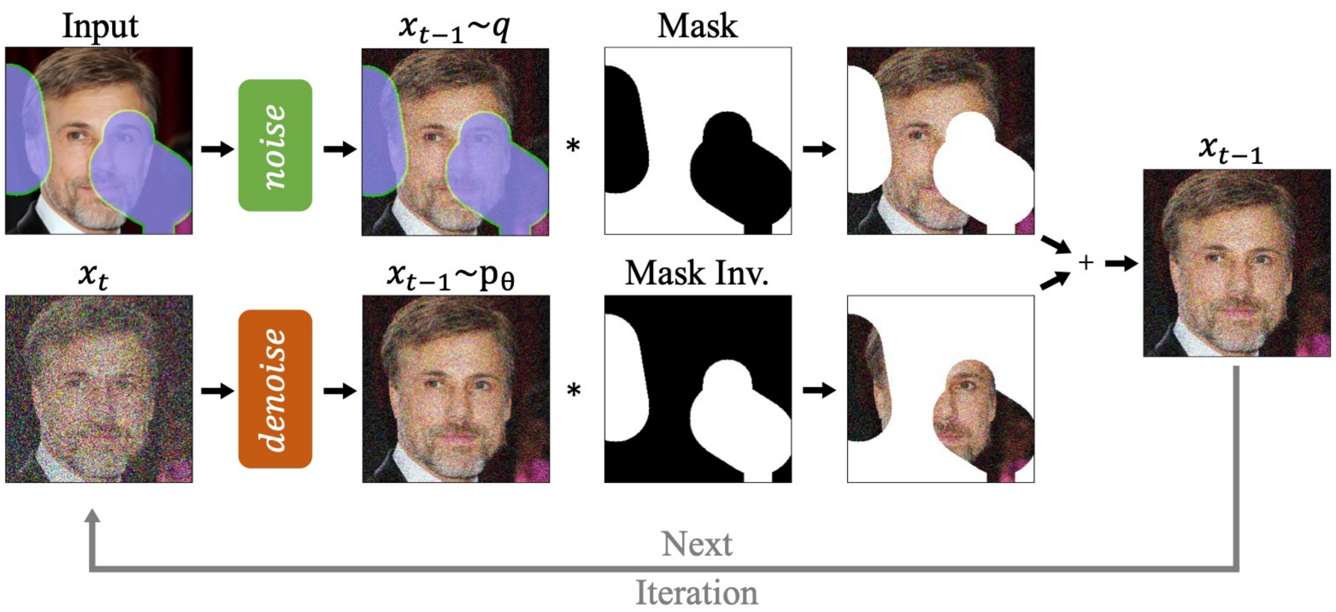
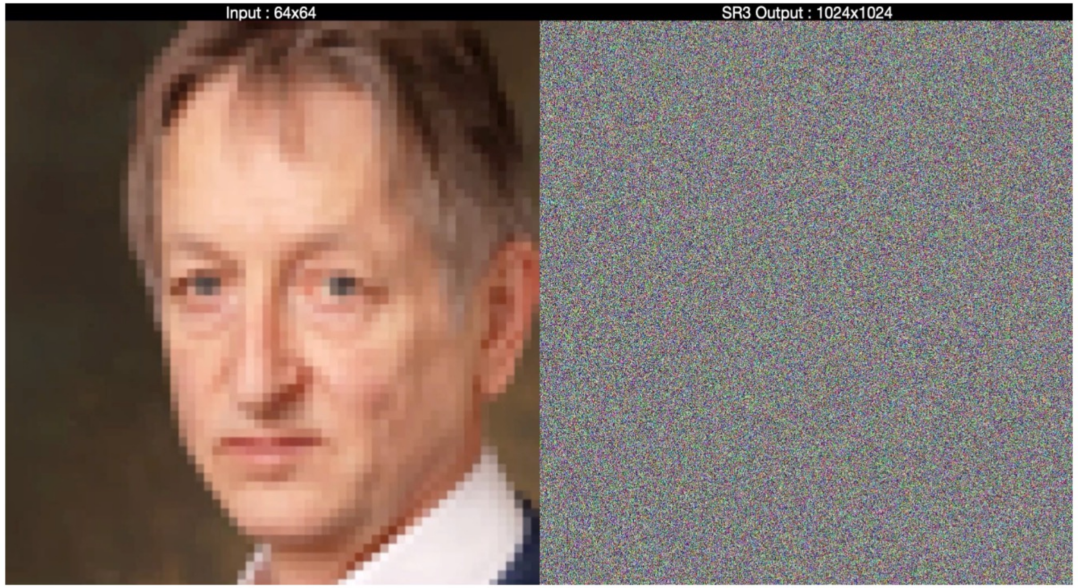


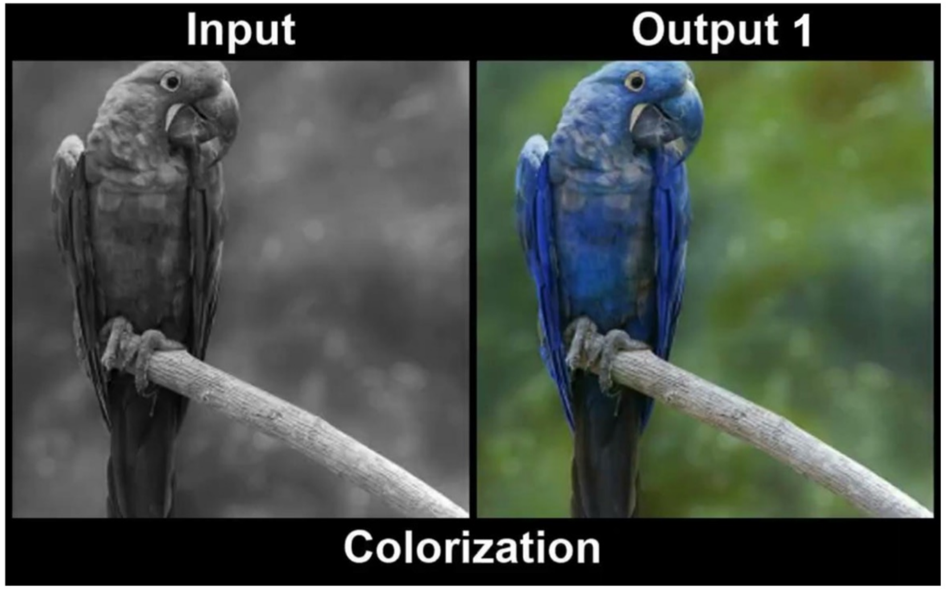
Figure 2. **Overview of our approach.** RePaint modifies the standard denoising process in order to condition on the given image content. In each step, we sample the known region (*top*) from the input and the inpainted part from the DDPM output (*bottom*).

[RePaint: Inpainting using Denoising Diffusion Probabilistic Models, Arxiv]()

Super-resolution & Colorization



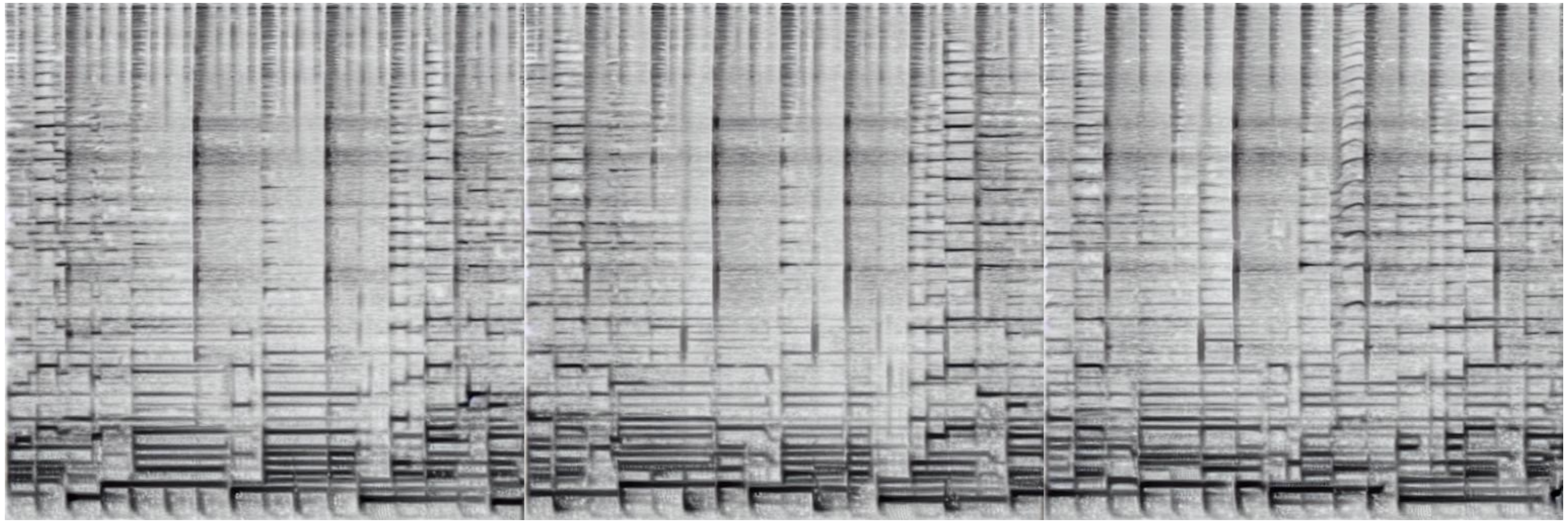
Super-resolution



Colorization

[Palette: Image-to-Image Diffusion Models, Arxiv]

Music Generation (Riffusion)



Riffusion, Wikipedia

Методы дообучения диффузионных моделей

Дообучение диффузионных моделей

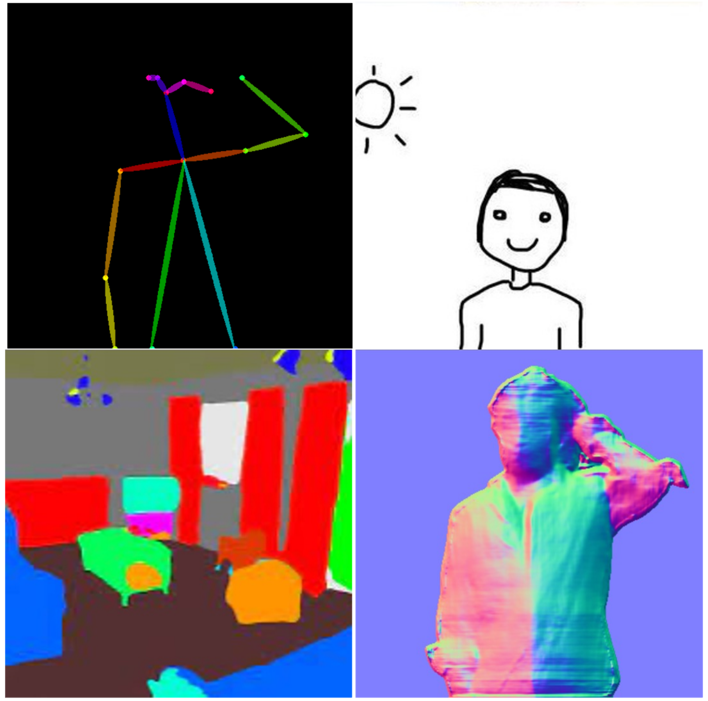
Стиль

Textual Inversion
Dreambooth

Объект

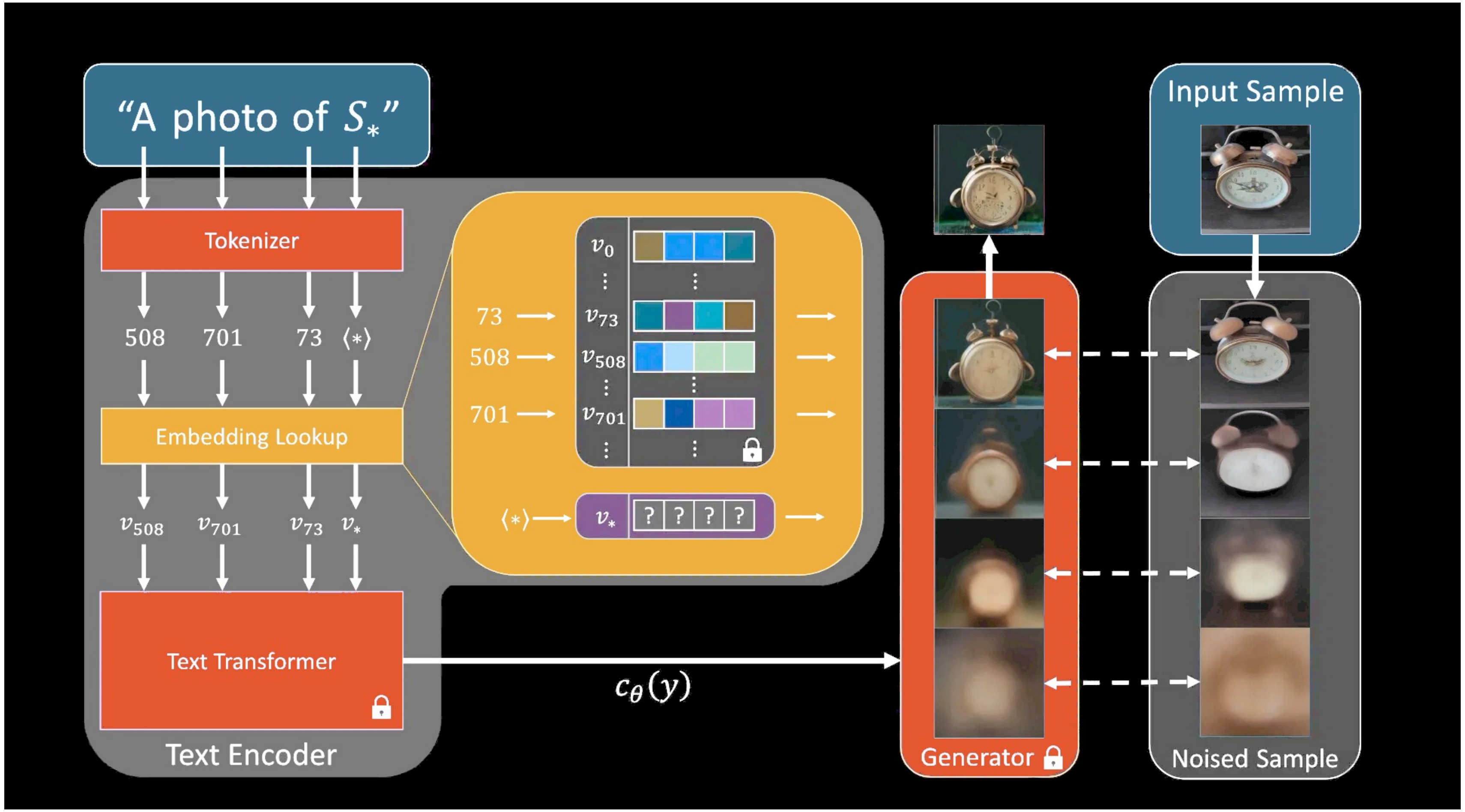
Textual Inversion
Dreambooth

Условие



Retraining
ControlNet
IP-Adapter

Textual Inversion

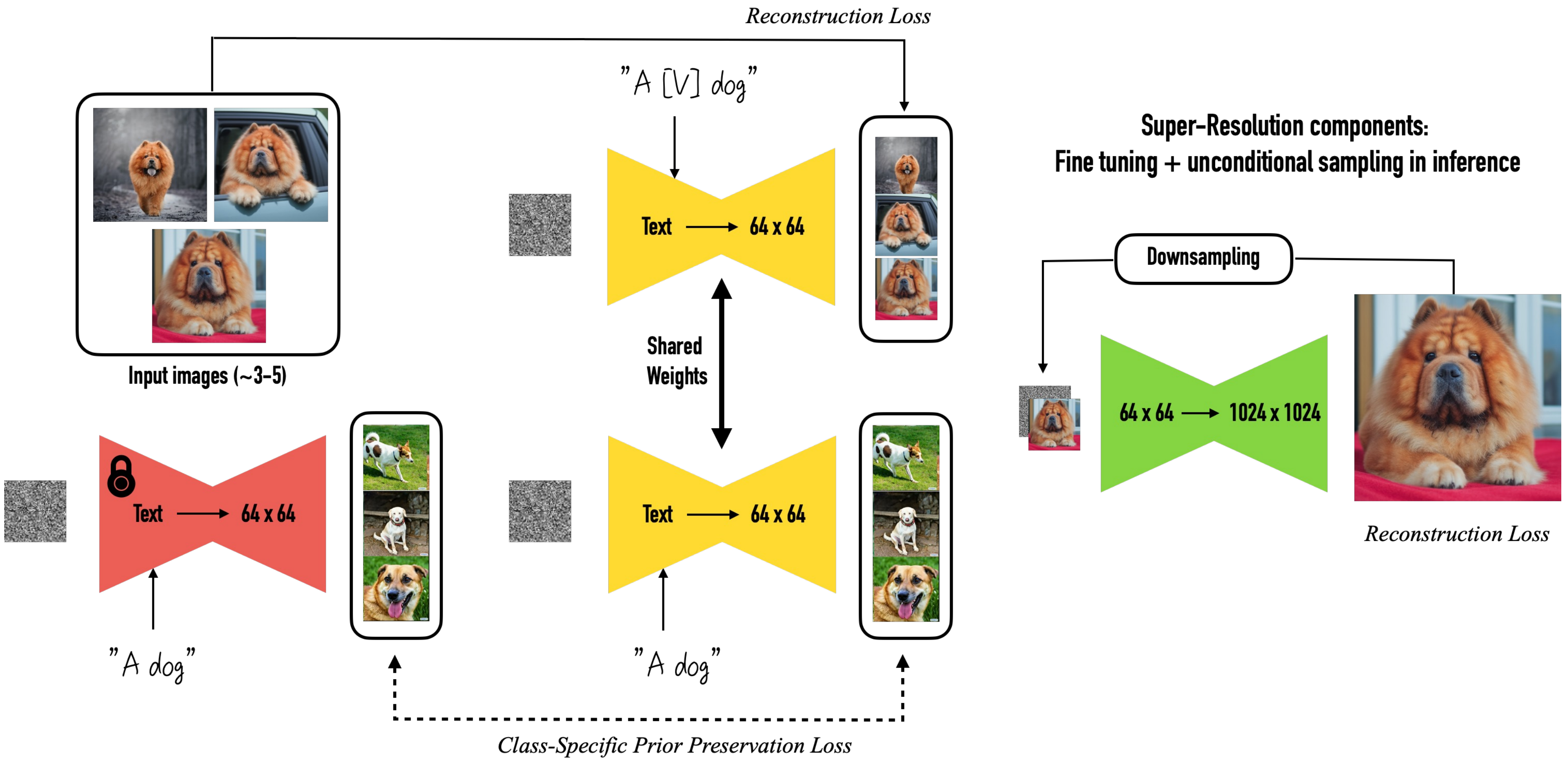


An Image is Worth One Word: Personalizing Text-to-Image Generation using Textual Inversion, Arxiv

Textual Inversion

[An Image is Worth One Word: Personalizing Text-to-Image Generation using Textual Inversion, Arxiv]()

Dreambooth



[DreamBooth: Fine Tuning Text-to-Image Diffusion Models for Subject-Driven Generation, Arxiv]()

Dreambooth

[DreamBooth: Fine Tuning Text-to-Image Diffusion Models for Subject-Driven Generation, Arxiv]

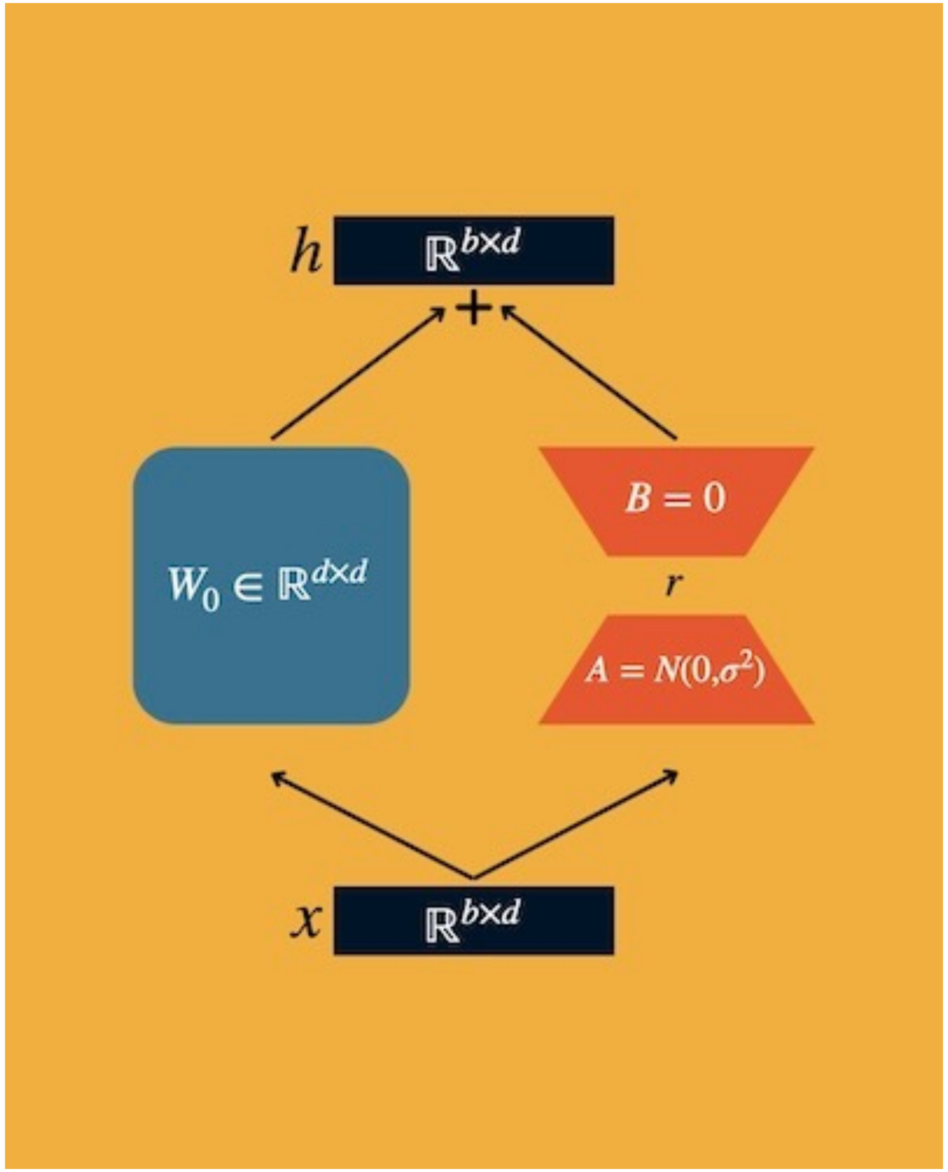
Dreambooth with LoRA

Снижаем требования к VRAM за счет обучения добавок $\Delta W = BA$ к слоям модели

$$h = W_0 x + \Delta W x = W_0 x + BAx$$

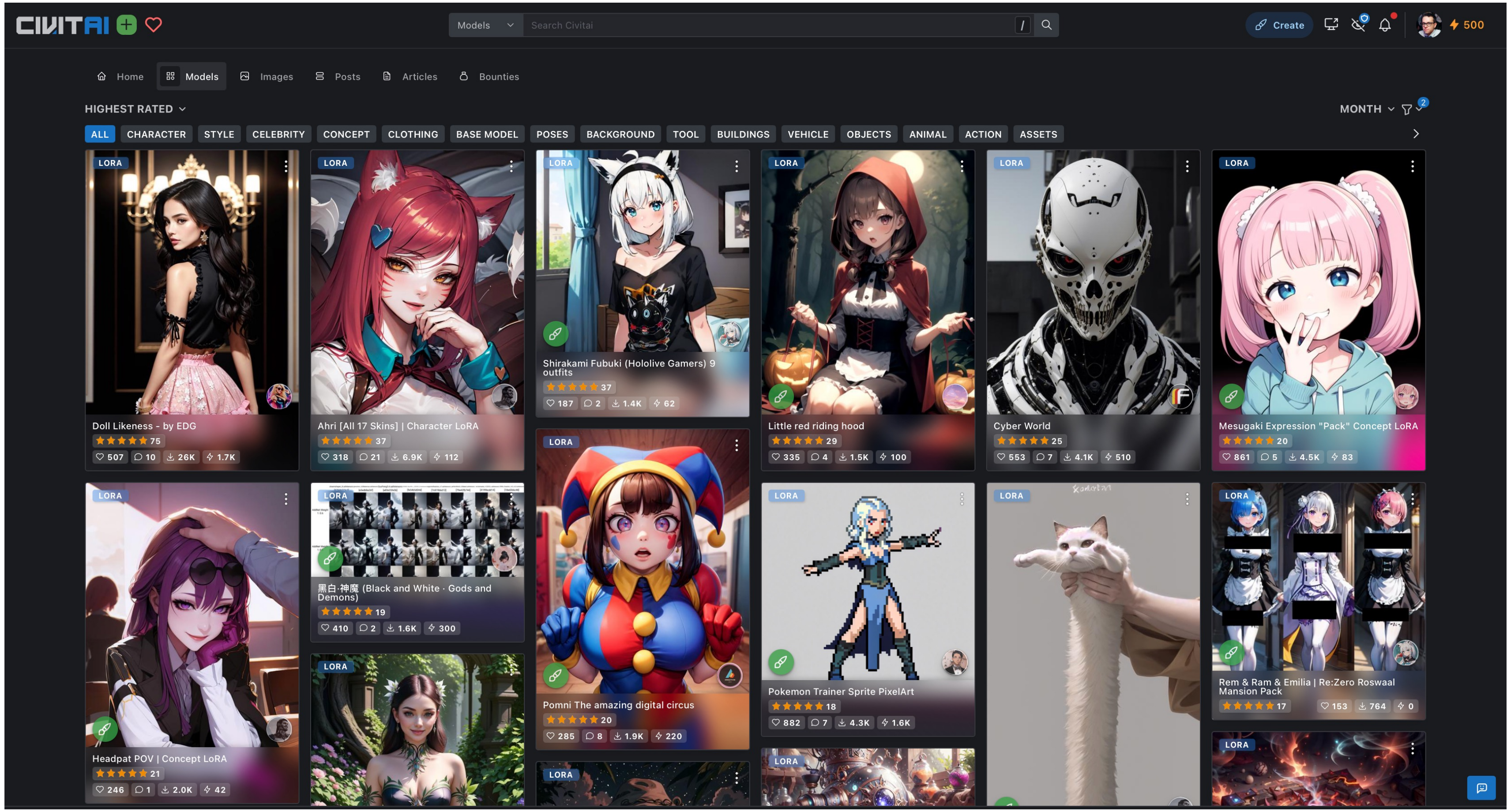
$$W \in R^{N\times M}, B \in R^{N\times r}, A \in R^{r\times M}$$

Получаем адаптер к весам базовой модели.



[LoRA: Low-Rank Adaptation of Large Language Models, Arxiv]

civitai.com



civitai.com

Смешивание LoRA

Методы добавления новых условий в обученные модели

ControlNet

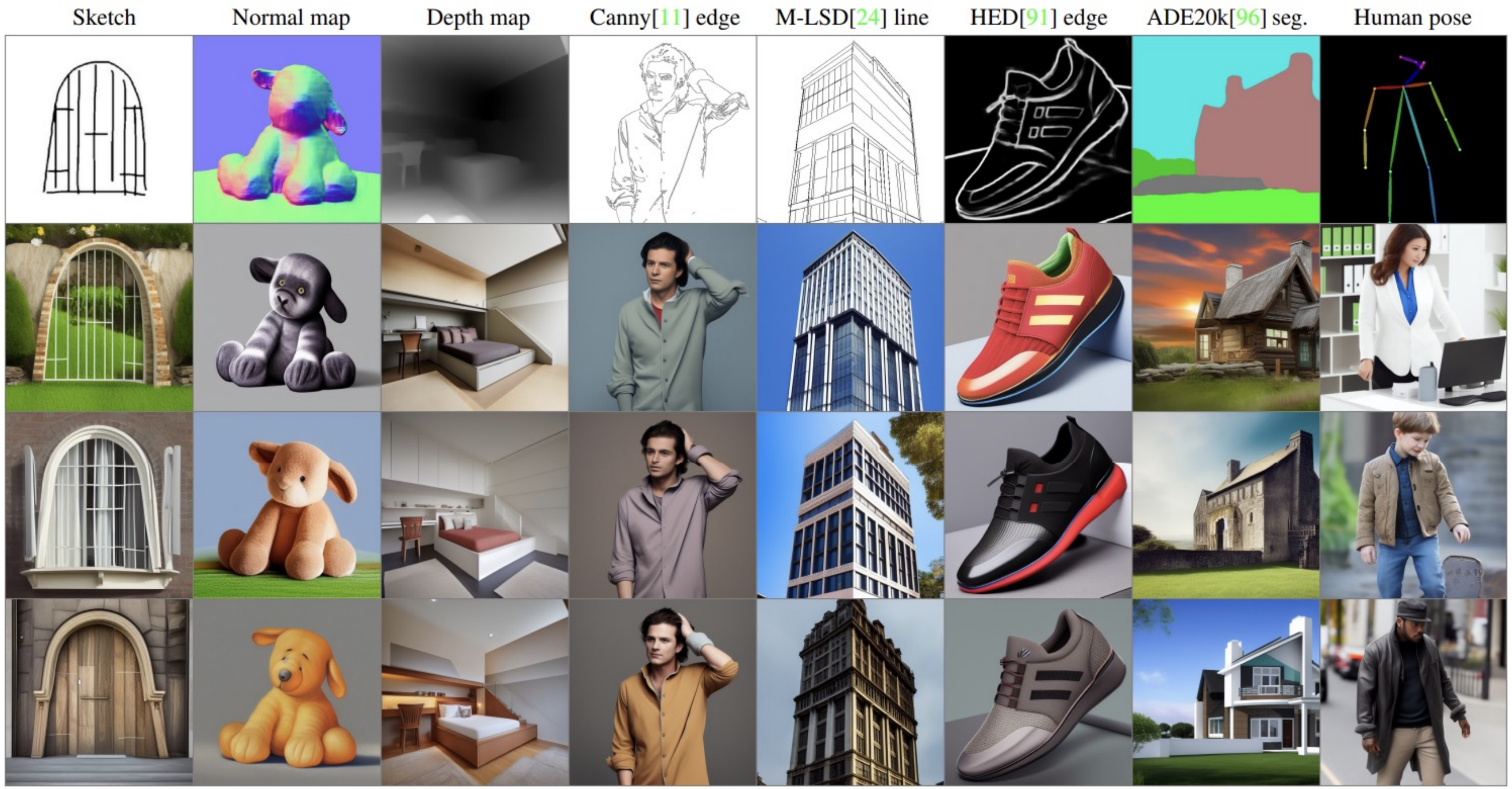
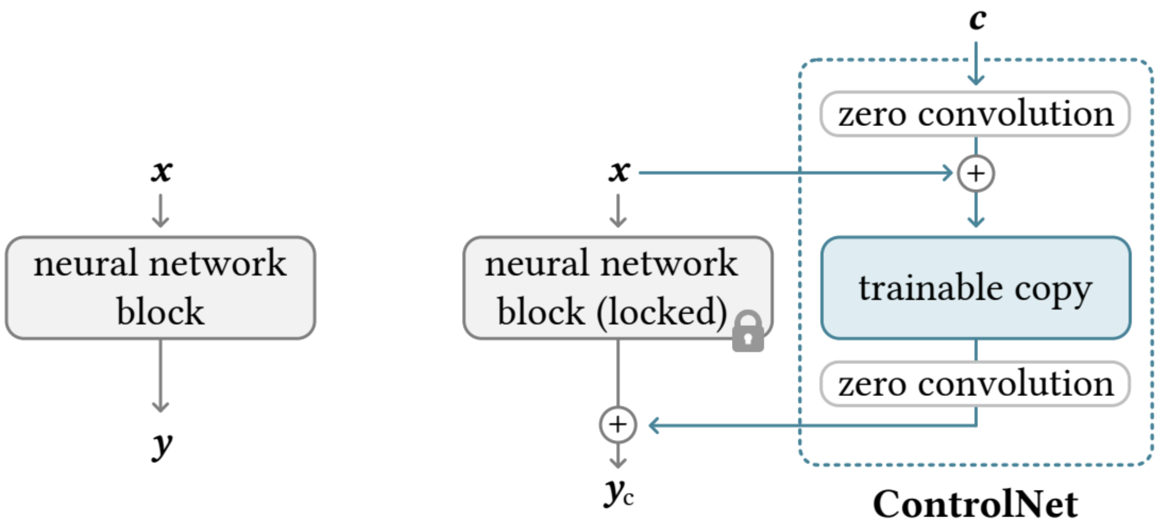


Figure 7: Controlling Stable Diffusion with various conditions **without prompts**. The top row is input conditions, while all other rows are outputs. We use the empty string as input prompts. All models are trained with general-domain data. The model has to recognize semantic contents in the input condition images to generate images.

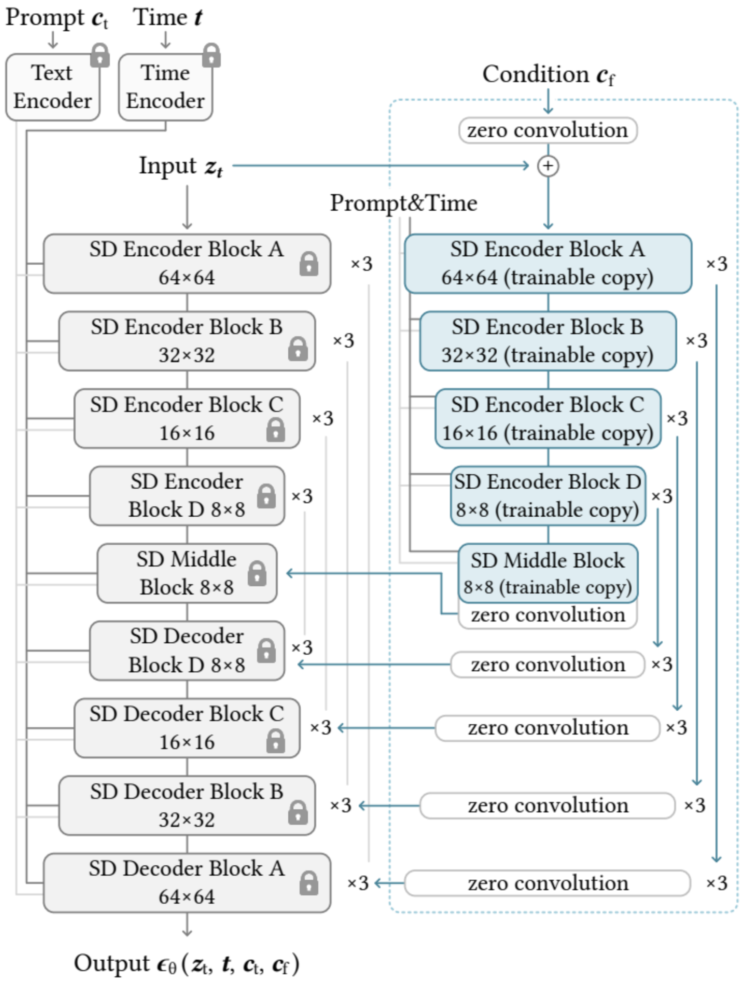
github.com/lllyasviel/ControlNet

ControlNet



(a) Before (b) After

Figure 2: A neural block takes a feature map x as input and outputs another feature map y, as shown in (a). To add a ControlNet to such a block we lock the original block and create a trainable copy and connect them together using zero convolution layers, *i.e.*, 1×1 convolution with both weight and bias initialized to zero. Here c is a conditioning vector that we wish to add to the network, as shown in (b).



(a) Stable Diffusion (b) ControlNet

Figure 3: Stable Diffusion's U-net architecture connected with a ControlNet on the encoder blocks and middle block. The locked, gray blocks show the structure of Stable Diffusion V1.5 (or V2.1, as they use the same U-net architecture). The trainable blue blocks and the white zero convolution layers are added to build a ControlNet.

[Adding Conditional Control to Text-to-Image Diffusion Models, Arxiv]

ControlNet

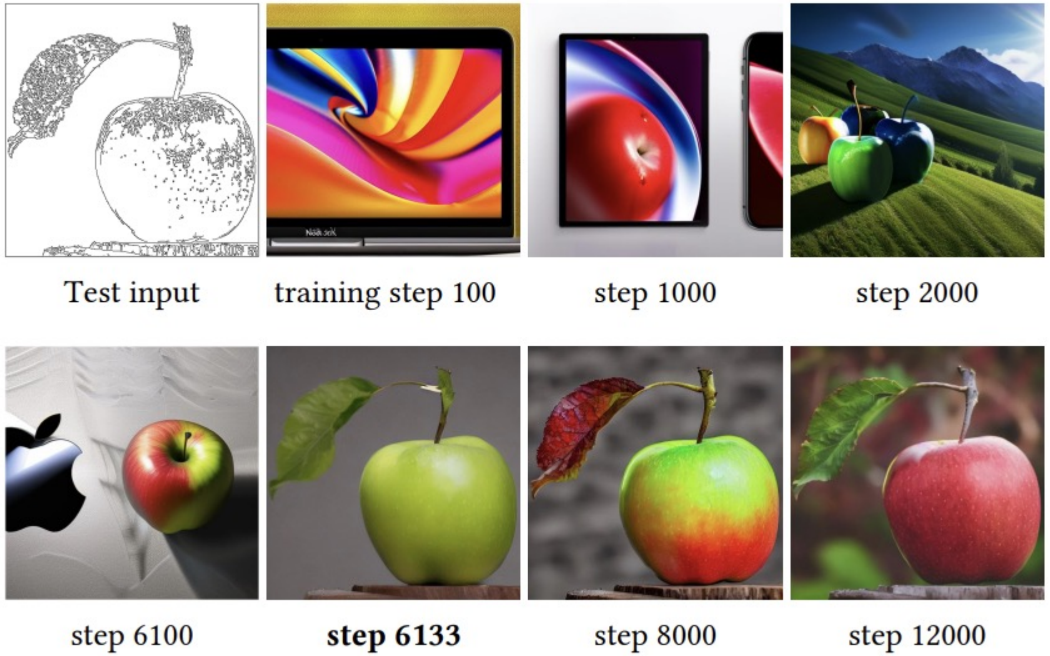


Figure 4: The sudden convergence phenomenon. Due to the zero convolutions, ControlNet always predicts high-quality images during the entire training. At a certain step in the training process (*e.g.*, the 6133 steps marked in bold), the model suddenly learns to follow the input condition.

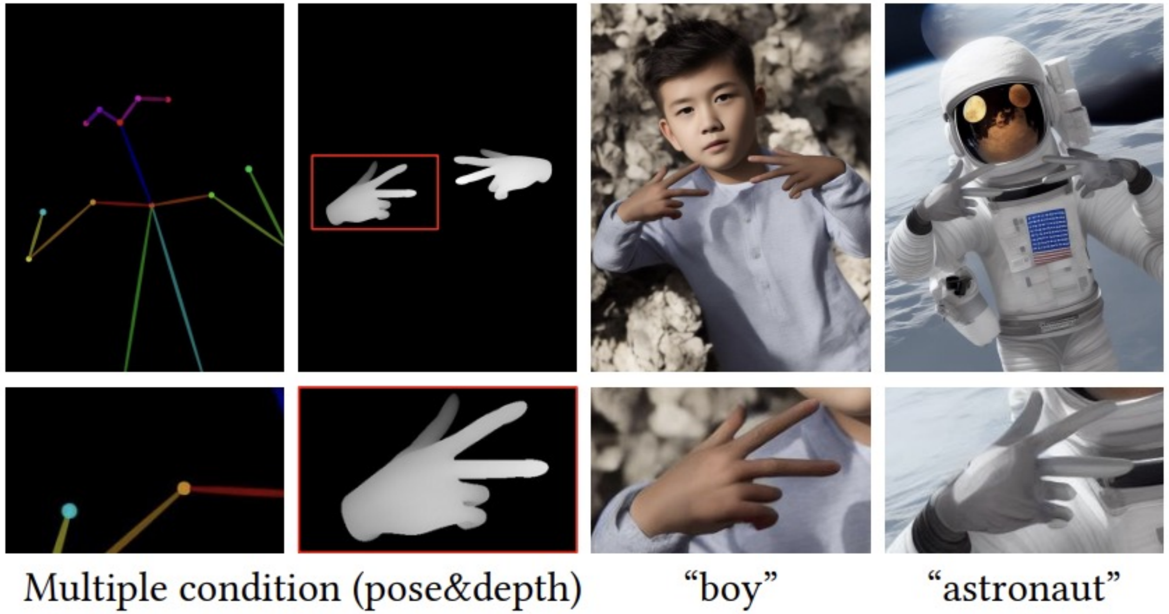


Figure 6: Composition of multiple conditions. We present the application to use depth and pose simultaneously.

[Adding Conditional Control to Text-to-Image Diffusion Models, Arxiv]

T2I-Adapter

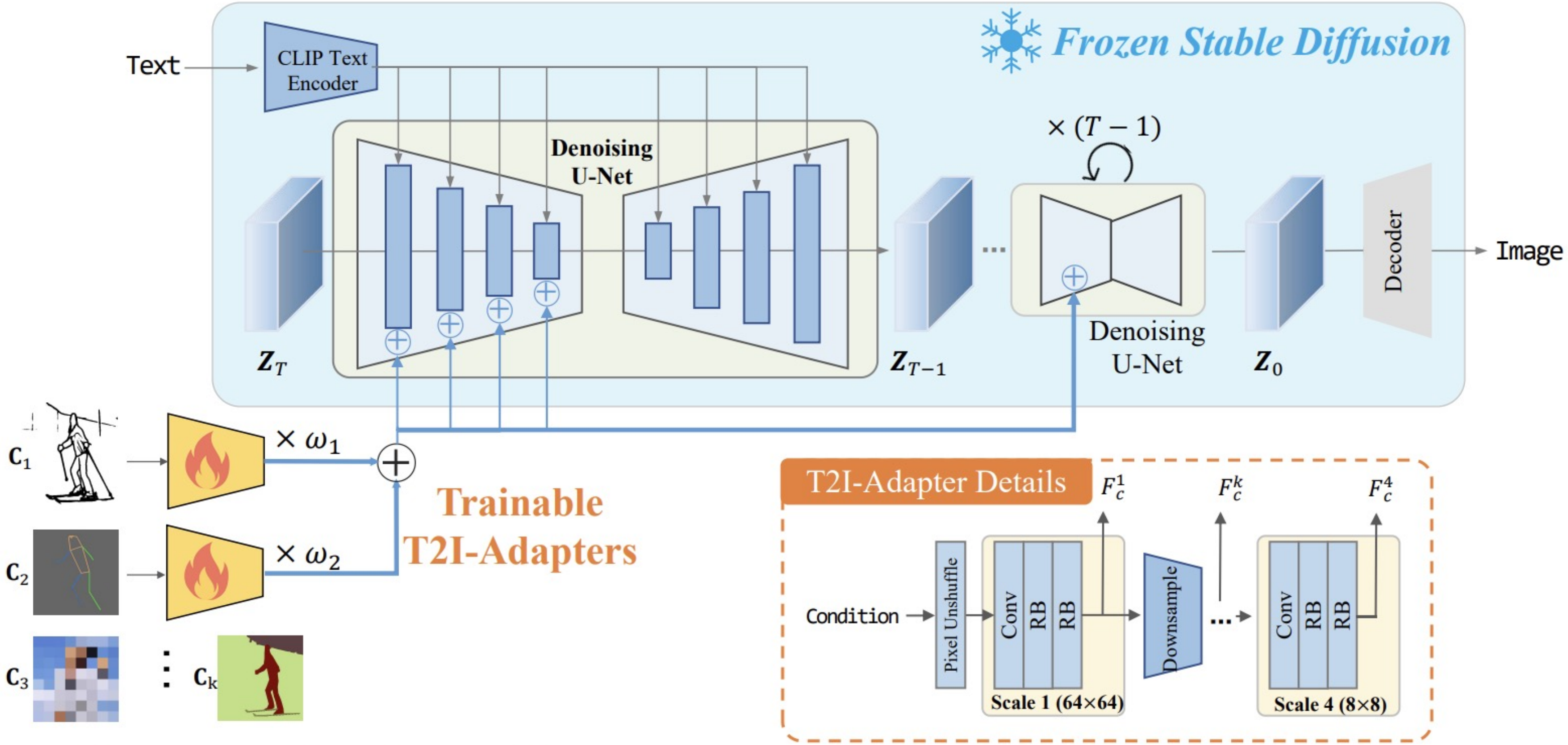


Figure 3. The overall architecture is composed of two parts: 1) a pre-trained stable diffusion model with fixed parameters; 2) several T2I-Adapters trained to align internal knowledge in T2I models and external control signals. Different adapters can be composed by directly adding with adjustable weight ω. The detailed architecture of T2I-Adapter is shown in the lower right corner.

[T2I-Adapter: Learning Adapters to Dig out More Controllable Ability for Text-to-Image Diffusion Models, Arxiv]

IP-Adapter

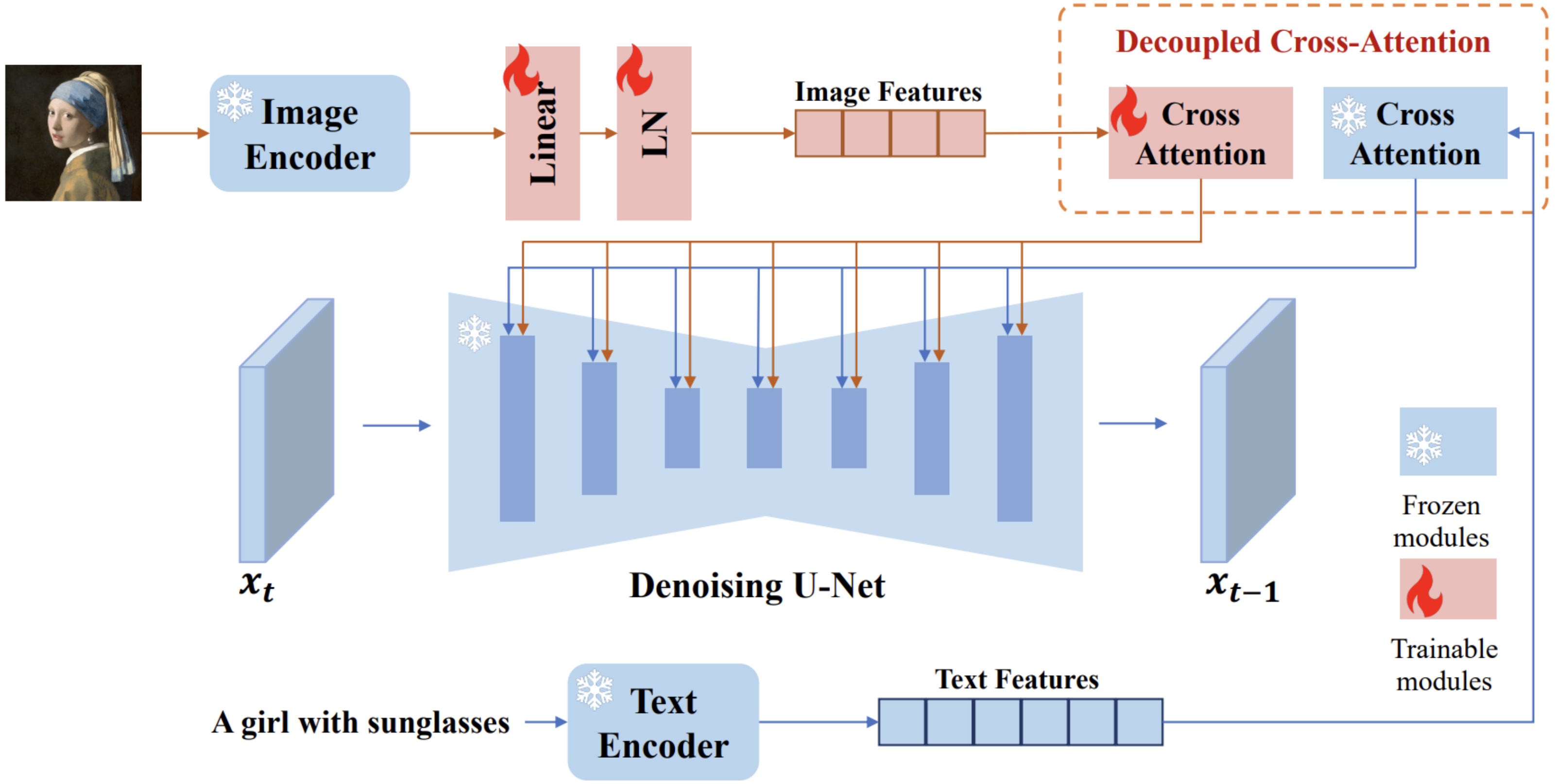


Figure 2: The overall architecture of our proposed IP-Adapter with decoupled cross-attention strategy. Only the newly added modules (in red color) are trained while the pretrained text-to-image model is frozen.

[IP-Adapter: Text Compatible Image Prompt Adapter for Text-to-Image Diffusion Models, Arxiv]()

IP-Adapter

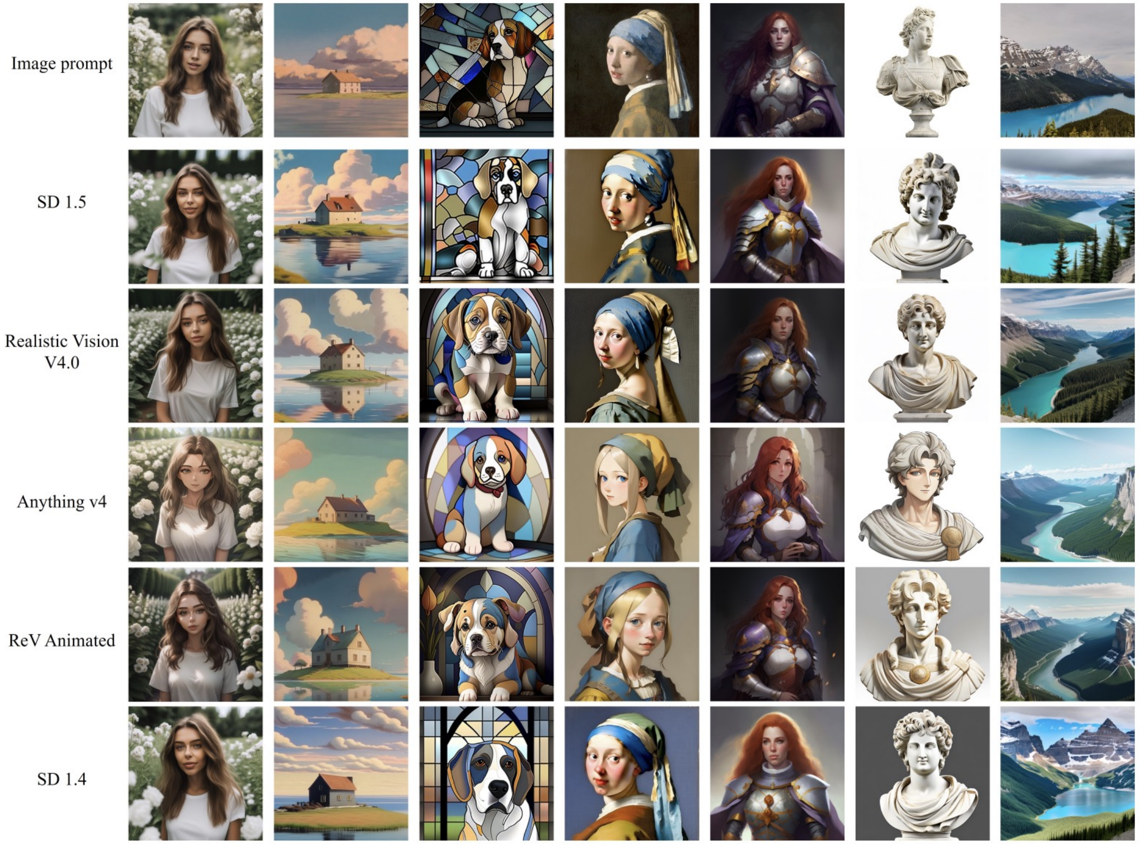
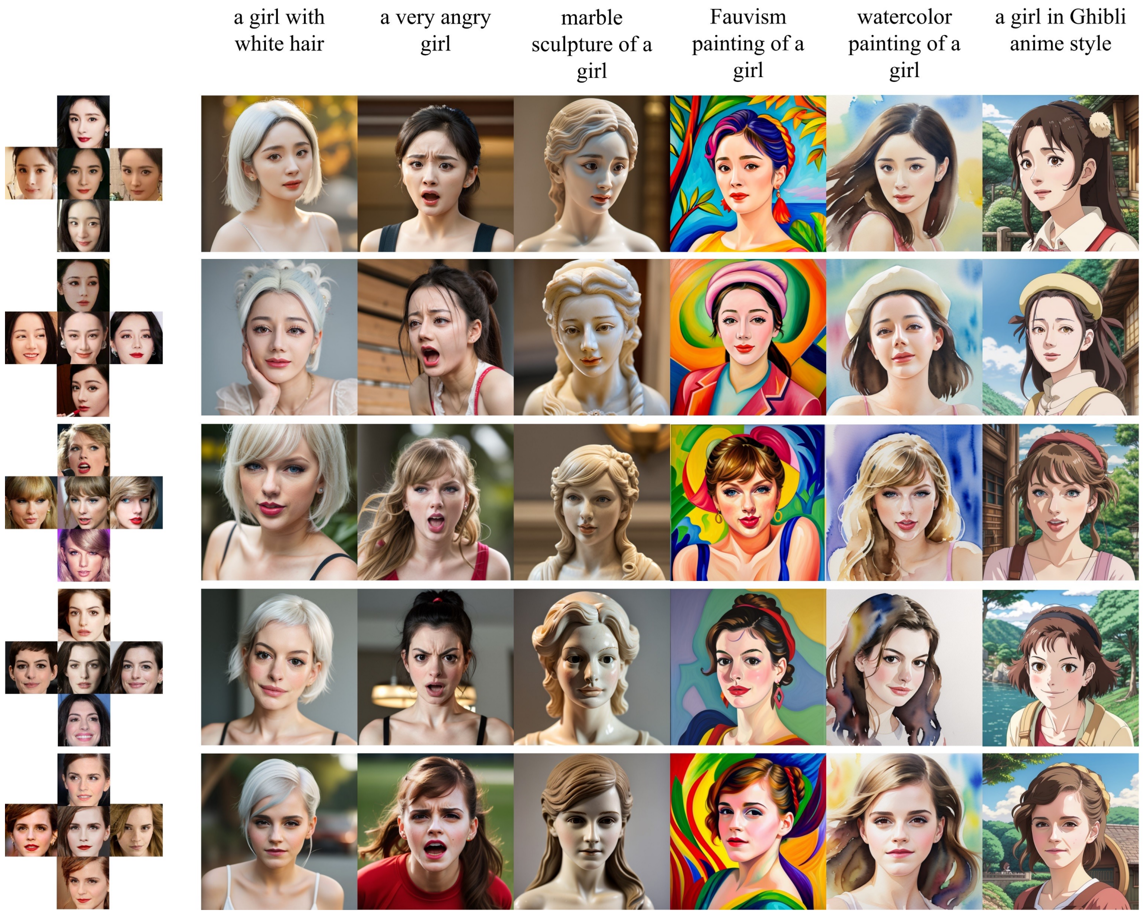


Figure 4: The generated images of different diffusion models with our proposed IP-Adapter. The IP-Adapter is only trained once.

[IP-Adapter: Text Compatible Image Prompt Adapter for Text-to-Image Diffusion Models, Arxiv]

IP-Adapter-FaceID-Portrait



https://huggingface.co/h94/IP-Adapter-FaceID

Собираем все вместе

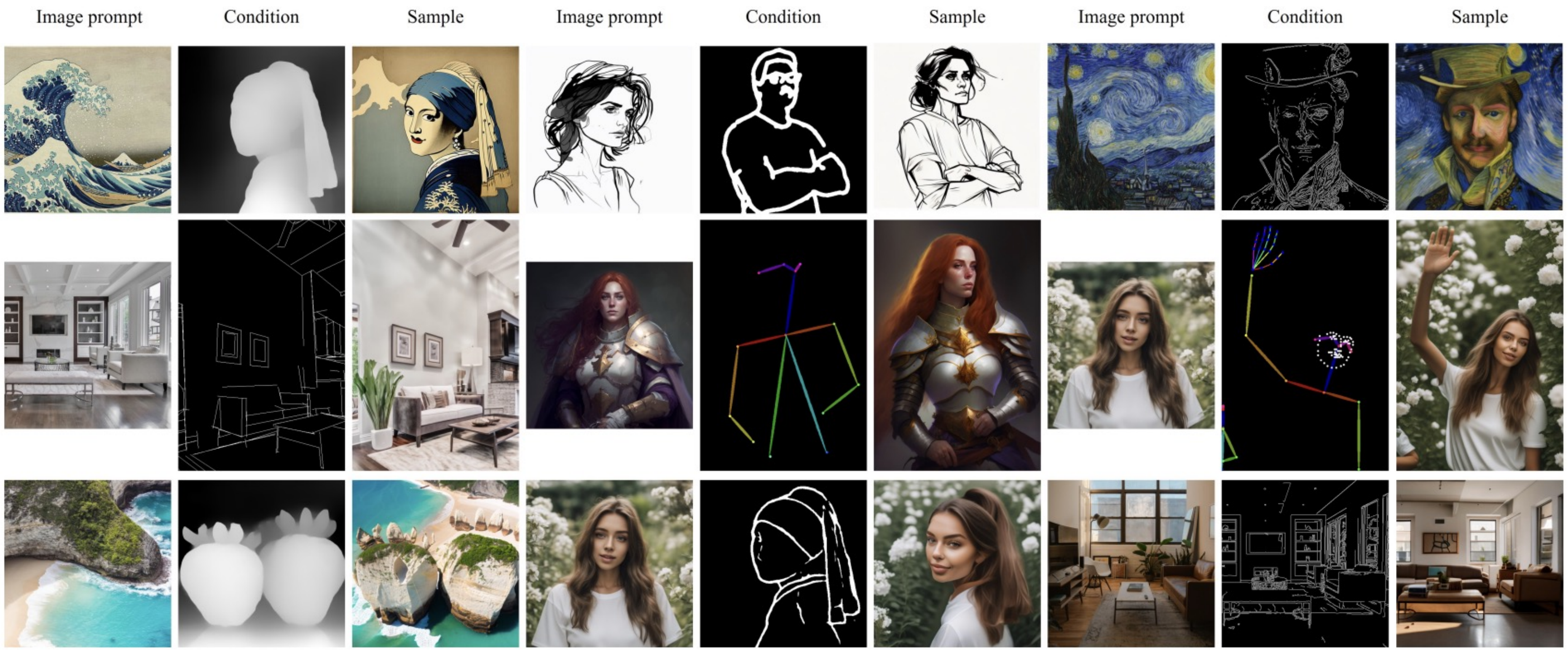
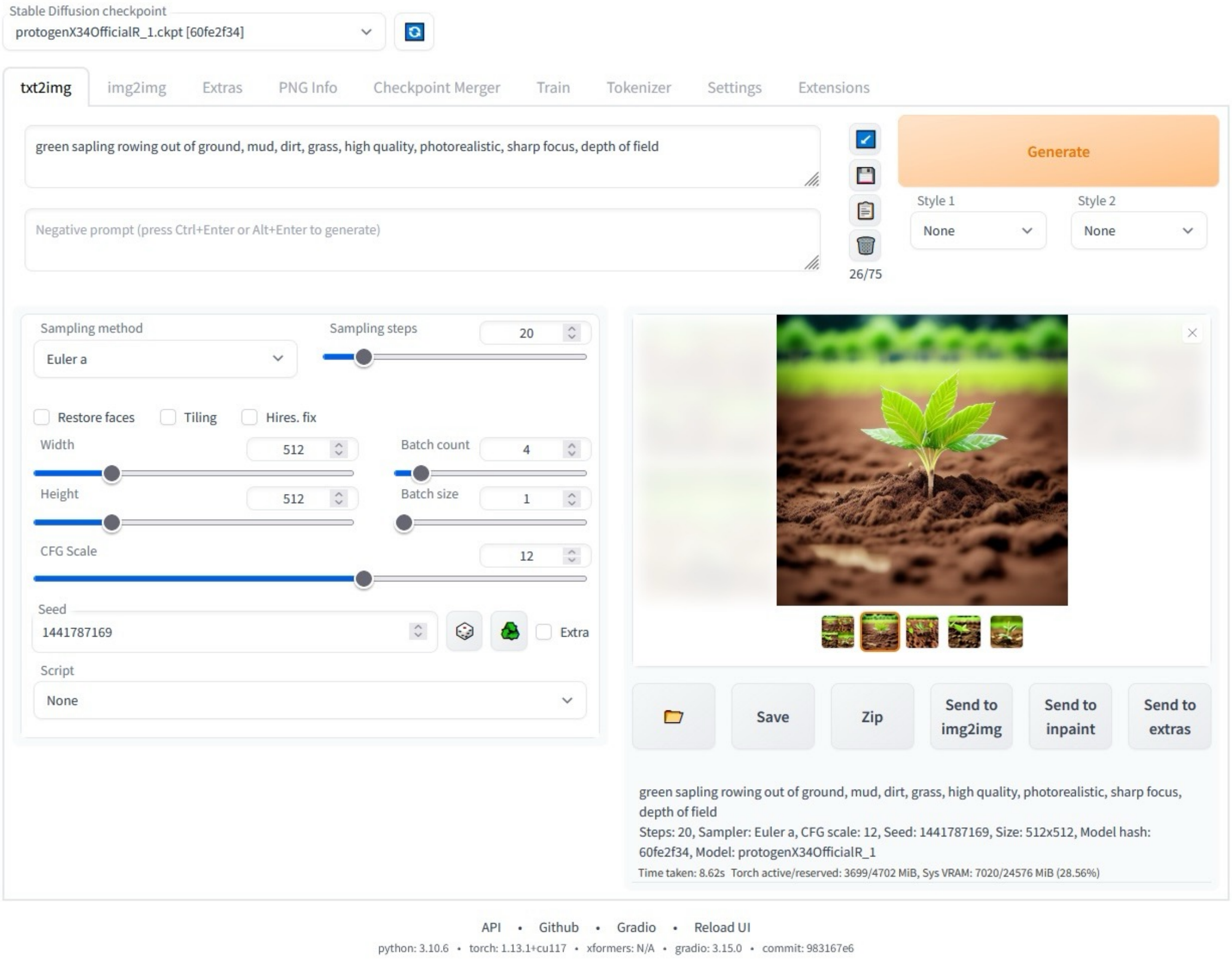


Figure 5: Visualization of generated samples with image prompt and additional structural conditions. Note that we don't need fine-tune the IP-Adapter.

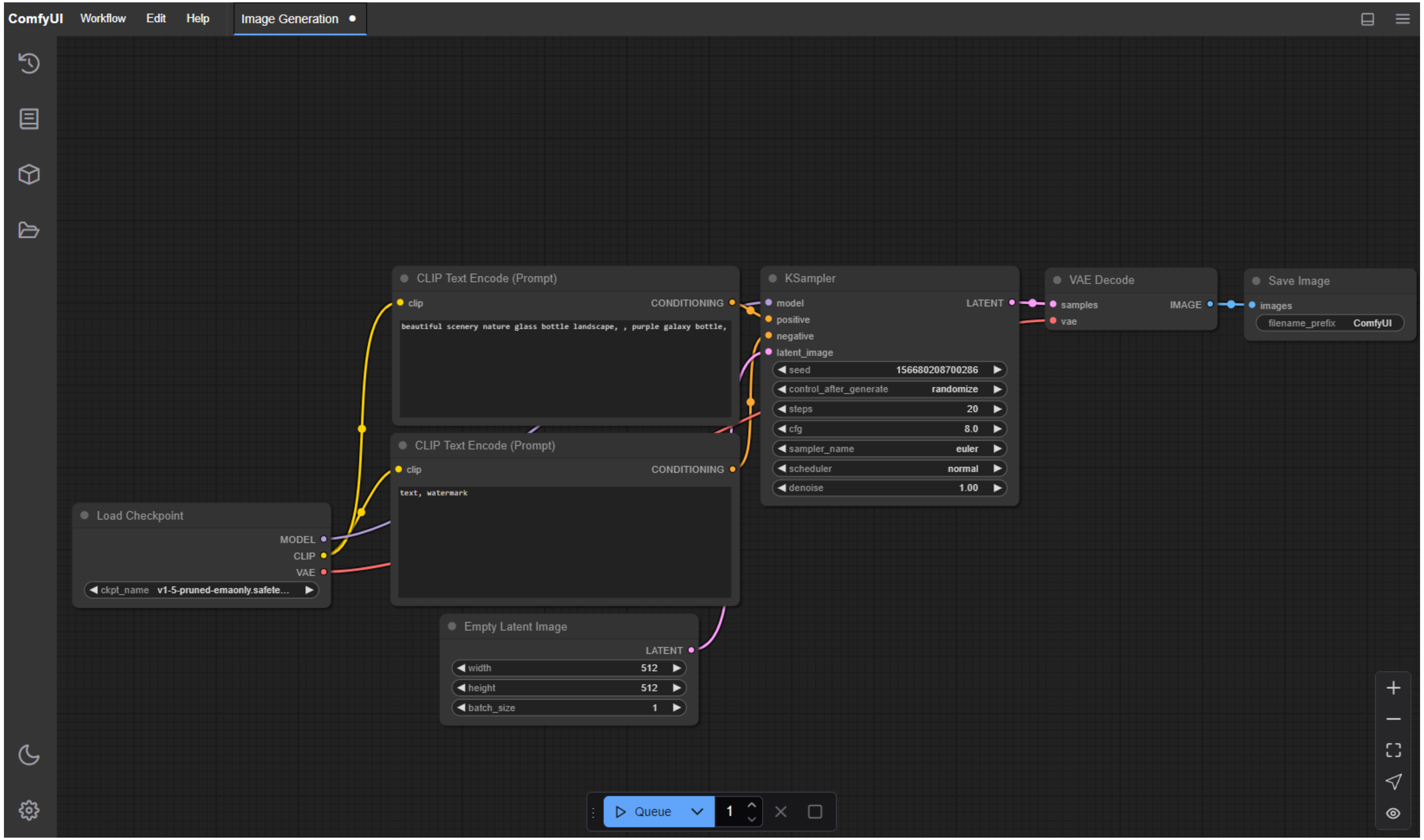
Инструменты для работы с диффузионными моделями

AUTOMATIC1111/stable-diffusion-webui



https://github.com/AUTOMATIC1111/stable-diffusion-webui

ComfyUI



https://github.com/comfyanonymous/ComfyUI

Заключение

- Рассмотрели основной концепт диффузионных моделей
- Поговорили об условной генерации и разных применениях диффузионных моделей
- Затронули тему дообучения моделей и добавления новых условий в уже существующие модели

Полезные ссылки

- Denoising Diffusion Implicit Models для ускорения сэмплирования
- Tutorial по диффузионным моделям на CVPR 2022 (взгляд на диффузионные модели с точки зрения дифференциальных уравнений)
- Блог Lillian Weng о диффузионных моделях
- Статья о Stable Diffusion 3 (DiT архитектура)